

Earnings Management Strategies and the Trade-Off between Tax Benefits and Detection Risk: To Conform or Not to Conform?

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Earnings Management Strategies and the Trade-Off between Tax Benefits and Detection Risk: To Conform or Not to Conform?

ABSTRACT: Prior research has separately examined pretax earnings management activities that have current taxable income consequences (book-tax ‘conforming earnings management’) and those that do not have current taxable income consequences (book-tax ‘nonconforming earnings management’). Our study documents the prevalence of, and then investigates the firm-specific characteristics that impact the choice between, these earnings management strategies. We utilize a sample of firms that restated their earnings downward due to accounting irregularities and thus can be presumed to have managed earnings upward. We find that nonconforming earnings management is more prevalent and that firms trade off the net present value of tax benefits against the net expected detection costs associated with nonconforming earnings management. In particular, firms having NOL carryforwards, using a high quality auditor, or engaging in the most egregious misstatements rely less on nonconforming earnings management strategies. We also find that book-tax differences are useful in predicting restatements.

Keywords: Earnings management; earnings restatements; book-tax differences; deferred tax expense; current tax expense.

Data Availability: All data used in this research are from publicly available sources.

I. INTRODUCTION

This study examines tax and nontax implications of pretax earnings management. A number of studies, e.g., Scholes et al. (1990), Matsunaga et al. (1992), Dhaliwal et al. (1994), and Erickson et al. (2004), investigate accounting choices and earnings management activities that have current taxable income consequences, which we refer to as book-tax *conforming* earnings management. Other studies, including Mills and Newberry (2001), Joos et al. (2003), and Phillips et al. (2003; 2004), consider earnings management that does not affect current taxable income (i.e., *nonconforming* earnings management). Utilizing a sample of firms that restated earnings due to accounting irregularities, we extend these prior studies in two ways. First, we document the prevalence of book-tax conforming and nonconforming earnings management in our sample of firms that restated their financial results (hereinafter, misstatement firms). Second, and more importantly, we investigate which firm-specific characteristics impact the choice between conforming and nonconforming earnings management strategies.¹

This research is important because it provides a better understanding of the choices firms make when managing earnings. By examining firms known to have misstated their reported earnings, we are able to estimate the relative prevalence of alternative earnings management strategies. Specifically, we examine the trade-offs managers face between the tax costs associated with pretax earnings management and the risks of being exposed as an earnings manager, and thus present evidence concerning the sensitivity of firms' earnings management choices to factors reflecting greater or lesser tax benefits and greater or lesser detection costs. Moreover, our research responds to Healy and Wahlen's (1999, 379) call for a better "understanding of the steps that firms take to increase reported earnings."

¹ We are concerned with pretax earnings management (i.e., managing revenue and expense accounts that affect pretax income) and how it is reflected in income tax accounts. Dhaliwal et al. (2004), Krull (2004), and Frank and Rego (2006) focus on managing the income tax accrual, which only affects after-tax earnings.

Our sample firms' misstatements reflect accounting irregularities and thus provide an interesting context in which to study earnings management (e.g., Richardson et al. 2002; Palmrose et al. 2004; Collins and Wan 2005; Badertscher et al. 2007, Hennes et al. 2007). We utilize the restatement setting to identify book-tax conforming and nonconforming earnings management because each choice has unique, predictable effects on the originally reported and restated current tax accounts. Consistent with Erickson et al. (2004), income-increasing conforming earnings management results in a downward restatement of pretax income and current tax expense. In contrast, income-increasing *nonconforming* earnings management results in a downward restatement of pretax income but not current tax expense.² Hence, restatements of current tax expense reveal the type of pretax earnings management (i.e., book-tax conforming or nonconforming) that managers employed.

In our preliminary analysis, we document that nonconforming earnings management accounts for the overwhelming proportion of misstatement firms' total pretax earnings management. Our primary analysis then investigates the trade-offs firms make between the tax benefits of nonconforming earnings management and the higher expected detection costs associated with such strategies.³ This framework builds on results in Mills (1998), Mills and Sansing (2000), and Hanlon and Krishnan (2005), which suggest that large book-tax differences trigger increased scrutiny by regulators and external auditors. We hypothesize that the lower the

² If the nonconforming earnings management generates a temporary book-tax difference, then the firm, consistent with Phillips et al. (2003), would restate deferred tax expense downward. If the nonconforming earnings management generates a permanent book-tax difference, then the firm would restate pretax income, but not current or deferred tax expense. Because we are interested in earnings management strategies that increase book income without affecting taxable income, we define nonconforming earnings management to include transactions that generate temporary *or* permanent book-tax differences. Nonetheless, supplemental analyses show that inferences from our empirical tests do not change if we focus on nonconforming earnings management that generates only temporary book-tax differences.

³ The "tax benefits of nonconforming earnings management" are equivalent to the tax costs of conforming earnings management. While conforming earnings management accelerates taxes relative to nonconforming earnings management, we generally refer to the "tax benefits of nonconforming earnings management" since we frame the manager's choice as trading off the costs and benefits of nonconforming strategies.

net present value of tax benefits or the higher the net expected detection costs associated with nonconforming earnings management, the less firms will rely on nonconforming strategies.

More specifically, we identify firm-specific characteristics that should impact the net present value of tax benefits or the net expected detection costs associated with nonconforming earnings management. First, we predict that firms with net operating loss (NOL) carryforwards have a lower net present value of tax benefits associated with nonconforming earnings management, and that firms with high free cash flow place a lower value on the tax benefits associated with nonconforming earnings management. Hence, we expect such firms to engage in more conforming earnings management. Second, we predict firms that use high quality external auditors, have greater analyst following, or engage in accounting fraud have higher expected detection costs associated with nonconforming earnings management and thus engage in more conforming earnings management.

Because we expect that using nonconforming earnings management strategies increases the chances of being detected as an earnings manager, we use the Heckman (1979) two-step estimation approach to help correct for potential self-selection bias in our misstatement sample. We first develop and estimate a prediction model for restatements, and then in the second step we estimate an earnings management strategy choice model after including the self-selection adjustment (the inverse Mills ratio) extracted from the first step. The results of the second-stage estimation are our main results and indicate, as predicted, that firms with lower tax costs, proxied by firms with NOL carryforwards, and firms with higher expected net detection costs, as proxied by firms that have a Big 4/5/6 auditor or that engage in the most egregious misstatements (e.g., accounting fraud), rely on significantly less nonconforming earnings management. In addition, we find evidence that firms with large cumulative temporary book-tax differences rely more on

conforming earnings management, suggesting that nonconforming earnings management in prior years constrains such firms in their use of nonconforming earnings management in the current period. Further, in contrast to conventional wisdom that revenue recognition is typically book-tax conforming, we find that firms that only manage revenue generally did so in a nonconforming manner by managing unearned revenue. Finally, a new result from the first-step of our Heckman estimation approach is that total book-tax differences have incremental explanatory power beyond abnormal accruals in predicting restatements.

Our research contributes to a better understanding of the tax and nontax factors that managers trade off when managing earnings upward. Erickson et al. (2004) document that some firms paid taxes in conjunction with income-increasing fraudulent earnings management. We extend their work by considering detection costs as well as the tax consequences of pretax earnings management and identify circumstances in which managers are likely to engage in more book-tax conforming earnings management. We also contribute to research that seeks to predict accounting irregularities, including fraudulent reporting (Richardson et al., 2002; Dechow et al. 2007), by demonstrating that total book-tax differences, and its major component, temporary differences, are incrementally useful in predicting restatements.

Our paper proceeds as follows. Section II develops our predictions and Section III discusses the research design. We describe our sample in Section IV and present the empirical results in Section V. Section VI concludes.

II. PRIOR RESEARCH AND PREDICTIONS

Background

We utilize data from financial statements and accompanying income tax footnotes to investigate the extent to which pretax earnings management affects current taxable income. Our

study builds upon several streams of research, including studies that investigate earnings management accomplished via book-tax conforming and nonconforming strategies and studies that consider earnings restatements as evidence of earnings management activity.

One stream of research that we build upon focuses on book-tax conforming upward earnings management, which increases both financial (i.e., book) and taxable incomes and thus has current income tax consequences (e.g., Scholes et al. 1990; Matsunaga et al. 1992; Dhaliwal et al. 1994; Erickson et al. 2004). Conforming earnings management includes any transaction that has the same impact on the current period's book and taxable incomes.⁴ Common examples of conforming earnings management include accelerating revenue recognition and recording receivables sooner than justified (e.g., revenue front-loading and channel-stuffing) and altering the timing of inventory purchases under LIFO.⁵ An example of a firm from our sample that primarily engaged in conforming earnings management is Versatility Inc., which determined that certain revenue had been inappropriately accelerated, the related receivables had been improperly recorded, and certain costs had not been accrued. We estimate that approximately 73 percent of Versatility's earnings management was conforming (see Appendix C).

In contrast, other studies focus on nonconforming upward earnings management, which increases book income but has no current tax consequences (e.g., Mills and Newberry 2001; Joos et al. 2003; Phillips et al. 2003; Phillips et al. 2004; also textbooks such as Revsine et al. 2005 and Palepu et al. 2003). This type of earnings management includes transactions that accelerate

⁴ Erickson et al. (2004) document that 15 of 27 firms that fraudulently overstated their financial accounting income paid, on average, 20 cents of current income tax for every dollar of overstated pretax income. Their results are consistent with some firms being willing to incur the tax costs associated with upward conforming earnings management to avoid the financial reporting costs of reporting lower earnings or to reduce the likelihood (*ex ante*) of detection. Our focus is not on cash taxes paid but instead considers how income tax accounts disclosed in firms' financial reports can be used to determine how firms manage their earnings.

⁵ Making purchases in the current year when input prices have fallen or postponing purchases at year-end when input prices have risen are examples of how earnings can be managed using LIFO.

revenue recognition or defer expense recognition for financial reporting purposes, relative to tax purposes, such as aggressively recognizing unearned revenue, extending the useful lives of depreciable assets, or reducing the provision for doubtful accounts for financial reporting purposes. These transactions, which have no effect on current taxable income, lead to more positive (or less negative) book-tax differences. An example of a firm from our sample that primarily engaged in nonconforming earnings management is Pre-Paid Legal Services Inc., which announced it had overstated revenue and understated expenses. We determine that approximately 97 percent of Pre-Paid Legal Services' earnings management was book-tax nonconforming (see Appendix C).⁶ Note in the above examples that both firms' misstatements involve several accounts and reflect a mix of book-tax conforming and nonconforming earnings management, suggesting that those firms had a portfolio of earnings management strategies available to them.

Mills and Newberry (2001) access confidential tax return data and use the total difference between a firm's book and taxable incomes as their proxy for nonconforming earnings management activity. Most book-tax difference studies, however, rely on publicly available data and use deferred tax expense or the change in net deferred tax liabilities as proxies for nonconforming earnings management activity.⁷ Since book-tax differences can reflect earnings management, tax planning, nondiscretionary book-tax rule differences, or other information about firms' economic activities (e.g., Hanlon et al. 2006; Dhaliwal et al. 2008), prior book-tax

⁶ This example is consistent with prior research that proposes, other things equal, firms should prefer to manage earnings upward in a manner that boosts book income relative to taxable income thereby minimizing current tax effects (e.g., Phillips et al. 2003).

⁷ Under Statement of Financial Accounting Standard No. 109, deferred tax expense reflects temporary differences between book and taxable incomes. See Phillips et al. (2003) for a more detailed discussion of SFAS No. 109.

difference studies likely measure nonconforming earnings management activity with error.⁸ Moreover, identifying settings in which earnings management is likely to have occurred is also problematic. For example, studies that focus on firms that just meet or beat earnings benchmarks (e.g., Burgstahler and Dichev 1997; Phillips et al. 2003; Durtschi and Easton 2005), or on firms with relatively high levels of debt (e.g., DeAngelo et al. 1994; DeFond and Jiambalvo 1994; Becker et al. 1998), not only rely on earnings management metrics that contain measurement error, but also lack certainty as to which firms engaged in earnings management.

We utilize a sample of firms that restated their financial results to correct previously reported accounting irregularities, and thus we are able to minimize error in identifying earnings management activity. As Badertscher et al. (2007, 20) note, these “accounting irregularities include aggressive accounting practices, material errors, misinterpretation of accounting rules, and intentional or unintentional misuse of facts and fraud.” Hence, studies utilizing such firms assume most restatements reflect earnings management (e.g., Richardson et al. 2002; Palmrose et al. 2004; Badertscher et al. 2007). Given uncertainty as to whether and/or to what extent, for example, benchmark beaters (e.g., Durtschi and Easton 2005) or firms with high debt levels manage earnings, restatements are a more reliable indicator of earnings management activity.⁹

Focusing on misstatement firms also enables us to use each firm as its own control and more accurately measure the amount of pretax earnings management (e.g., Erickson et al. 2004). The misstatement sample provides measures of managed financial results (i.e., originally

⁸ Likewise, studies that use abnormal accruals to measure earnings management also do so with error (e.g., Guay et al. 1996).

⁹ Although we analyze earnings management strategies for a sample of firms likely to have managed earnings, we cannot claim with certainty that all of the earnings management was intentional. An even more extreme sample would be that of firms known to have committed accounting fraud. While there is no uncertainty in such a sample about the occurrence of egregious earnings management, such a sample would be smaller than our sample of misstatement firms. We test fraud firms versus non-fraud firms, and thus provide insight as to whether our results are restricted to fraud firms or generalize to the larger set of misstatement firms.

reported earnings and its components) and unmanaged financial results (i.e., restated earnings and its components) for each misstatement firm. We hand-collect such data, which enables us to measure total pretax earnings management as the difference between restated and original pretax income.¹⁰ Moreover, the misstatement setting is also attractive because it is difficult, if not impossible, to control for the amount of tax planning that firms engage in. By comparing originally reported and restated data for the same firm and year, we are able to hold constant the effects of a firm's tax planning on current and deferred income tax expense. For instance, we can interpret changes in current tax expense resulting from restatements as being related to conforming earnings management activity and unrelated to tax planning activity. Similarly, we are also able to hold constant the effects of other items that impact the tax accounts (e.g., tax credits and tax benefits from stock option exercises) but that are not affected by the restatement. In sum, restatements allow us to more accurately identify a setting in which earnings management is very likely to have occurred, to better measure the total amount of managed earnings, and to classify the earnings management strategies (i.e., conforming or nonconforming) relative to prior earnings management research.

Predictions

Previous research documents conforming upward earnings management in certain contexts (e.g., LIFO abandonments (Johnson and Dhaliwal 1988) and fraudulent earnings overstatements (Erickson et al. 2004)¹¹), and nonconforming upward earnings management in other contexts (e.g., earnings benchmarks (Mills and Newberry 2001 and Phillips et al. 2003)). Hence, prior research has considered conforming and nonconforming earnings management

¹⁰ Restated financial results can presumably still reflect unacknowledged earnings management; nonetheless, we assume that restated amounts reflect strictly less earnings management than the amounts firms originally reported.

¹¹ Erikson et al. (2004) note that many firms in their sample deferred some taxes on earnings overstatements.

separately, i.e., in different settings in which one strategy or the other was more likely to be used. We analyze a setting where, because there is a high probability that earnings management occurred, we can examine the extent firms opted to use conforming or nonconforming methods. Thus, we initially investigate whether our sample of misstatement firms engaged in more nonconforming than conforming upward earnings management.

Our main focus, however, is to directly investigate the trade-off between the tax benefits and the higher risk of detection associated with nonconforming earnings management. We do so by considering conditioning circumstances in which, other things equal, we expect that firms are more likely to choose to manage earnings upward in a book-tax *conforming* manner. We assume firms have sufficient opportunities to engage in both conforming and nonconforming earnings management,¹² and propose that firms manage earnings in a nonconforming manner when the net present value of tax benefits (Net PVTB_{NC}) exceeds the net expected detection costs associated with nonconforming earnings management (Net E[DC]_{NC}); i.e., when:

$$\text{Net PVTB}_{\text{NC}} > \text{Net E[DC]}_{\text{NC}}. \quad (1)$$

We define the net present value of tax benefits from nonconforming earnings management as the present value of the tax costs related to book-tax conforming earnings management (PVT_C) minus the present value of the tax costs related to *nonconforming* earnings management (PVT_{NC}). We define net expected detection costs as the expected detection costs related to *nonconforming* earnings management (E[DC_{NC}]) minus the expected detection costs related to conforming earnings management (E[DC_C]). Accordingly, firms will engage in *nonconforming* earnings management when:

¹² In our empirical analysis, we control for a firm's earnings management opportunity set, including its opportunities to employ conforming and nonconforming methods. If firms do not have both conforming and nonconforming methods available to manage earnings, then our multivariate analysis of the trade-offs of tax benefits and detection costs will not yield significant results for the tax benefit and detection cost variables.

$$PVTC_C - PVTC_{NC} > E[DC_{NC}] - E[DC_C]. \quad (2)$$

We define the present value of tax costs as the amount of pretax earnings management (PT_EM) times the statutory tax rate (STR), and in the case of nonconforming earnings management this amount is discounted by a factor that increases in both the discount rate (r) and the number of years (n) it would take the nonconforming earnings management to reverse $((1+r)^n)$.¹³ Appendix A shows that the net present value of tax benefits associated with nonconforming earnings management ($Net\ PVTB_{NC}$) is increasing in PT_EM , STR , r , and n .

We define expected detection costs ($E[DC]$) as the probability of detection ($Pr(D_{NC})$ or $Pr(D_C)$) multiplied by total detection costs (DC). Examples of the possible costs of being detected as an earnings manager include Securities and Exchange Commission investigations and Internal Revenue Service audits, the related costs of appeals and penalties resulting from such investigations and audits, and negative share price effects and higher costs of capital associated with restatements.

Thus, considering the tax benefits and detection costs of both types of earnings management strategies, firms will engage in nonconforming earnings management when:

$$\{PT_EM \times STR - [(PT_EM \times STR) / (1+r)^n]\} > \{[Pr(D_{NC}) \times DC] - [Pr(D_C) \times DC]\}. \quad (3)$$

Rearranging terms leads to the following expression:

$$PT_EM \times STR \{1 - [1 / (1+r)^n]\} > [Pr(D_{NC}) - Pr(D_C)] \times DC, \quad (4)$$

where we assume that total detection costs (DC) do not vary with the type of earnings management strategy, and that $[Pr(D_{NC}) - Pr(D_C)] > 0$. This last assumption, that the probability of detection is greater for nonconforming than conforming earnings management, is based on prior research that argues book-tax differences often attract the attention of regulators such as the

¹³ We do not have to assume that all nonconforming earning management reverses. To incorporate non-reversing accruals, we assume that $n \rightarrow \infty$, which could also represent permanently nonconforming earnings management.

IRS (Mills 1998; Mills and Sansing 2000) and the SEC (Frank et al. 2008). Further, Hanlon and Krishnan (2005) find that external auditors increase the time and effort they spend on clients that have larger book-tax differences. These studies suggest that nonconforming relative to conforming earnings management should lead to a higher probability of detection and, in turn, higher net expected detection costs.

Equations (1) – (4) motivate two hypotheses; the first hypothesis focuses on the net present value of tax benefits associated with nonconforming earnings management (H1), and the second focuses on the net expected detection costs associated with nonconforming earnings management (H2). These hypotheses are (stated in the alternative):

H1: Other things equal, the lower the net present value of tax benefits associated with nonconforming earnings management, the lower the rate of nonconforming earnings management.

H2: Other things equal, the greater the net expected detection costs associated with nonconforming earnings management, the lower the rate of nonconforming earnings management.

We develop proxies for the net present value of tax benefits and net expected detection costs associated with nonconforming earnings management in the next section.

III. RESEARCH DESIGN

Prevalence of Conforming and Nonconforming Earnings Management

We use the difference between originally reported and restated current tax expense disclosed in earnings restatements to measure the amount of book-tax conforming earnings management (e.g., Erickson et al. 2004). An advantage of this research design is that it avoids much of the measurement error inherent in using current tax expense to estimate taxable income (e.g., Hanlon 2003) because we are using each firm as its own control. That is, isolating restatement-induced *changes* in current tax expense holds the effect of tax credits and other such

items constant, thus enabling us to measure the resulting *changes* in current taxable income attributable to conforming earnings management with less error than is typically associated with measuring taxable income using financial statement data. However, NOL carryforwards add measurement error to this proxy for conforming earnings management. In particular, if firms that manage earnings upward in a book-tax conforming manner have NOL carryforwards to offset the increase in taxable income, then originally reported and restated current tax expense will measure conforming earnings management with error.¹⁴ In addition, multinational corporations can manage U.S. and/or foreign pretax income. Thus, changes in current tax expense can be attributable to differences between originally reported and restated U.S. and/or foreign earnings. To address these issues, we use hand-collected financial report data to compute the amount of conforming earnings management as follows:

$$CONFORM_EM = \frac{(\Delta FED_CTE - \Delta FEDNOL_DTA)}{FED_RATE} + \frac{(\Delta FOR_CTE - \Delta FORNOL_DTA)}{FOR_RATE} \quad (5)$$

where the amount of conforming earnings management, *CONFORM_EM*, is based on current federal and foreign tax expense adjusted for changes in a firm's deferred tax assets resulting from NOL carryforwards. We gross-up the adjusted changes in federal and foreign current tax expense (i.e., $(\Delta FED_CTE - \Delta FEDNOL_DTA)$ and $(\Delta FOR_CTE - \Delta FORNOL_DTA)$) by, respectively, the federal statutory tax rate (*FED_RATE*) and the estimated foreign tax rate

¹⁴ Appendix B illustrates NOL-induced measurement error.

(*FOR_RATE*), so that *CONFORM_EM* is an estimate of the total dollar amount of conforming earnings management.¹⁵

INSERT FIGURE 1 HERE

We scale *CONFORM_EM* by the total amount of pretax earnings management (i.e., *PT_EM*) to compute the proportion (or rate) of total earnings management that was book-tax conforming (*C_RATE*).¹⁶ We then compute the proportion of total earnings management that was book-tax nonconforming (*NC_RATE*) as 1 minus *C_RATE*.¹⁷ Appendix C illustrates the calculation of *C_RATE* and *NC_RATE* for two firms in our sample: one that uses predominantly conforming methods and one that uses predominantly nonconforming methods. In our preliminary analysis, we compare sample means and medians to determine whether *NC_RATE* is significantly greater than *C_RATE*.

The Choice between Conforming and Nonconforming Earnings Management

¹⁵ We found that firms disclosed deferred tax assets related to NOL carryforwards more frequently than the dollar amounts of NOL carryforwards; thus, we utilize the total amount of deferred tax assets related to NOL carryforwards to adjust current tax expense. We allocate the total amount of deferred tax assets related to NOL carryforwards (*ANOL_DTA*) between the U.S. and foreign tax jurisdictions by multiplying *ANOL_DTA* by the ratios of U.S. or foreign pretax incomes to total pretax income, respectively. Also, because firms often operate in multiple foreign jurisdictions, we estimate the foreign tax rate that applies to any restated foreign earnings as the change in foreign current tax expense (*ΔFOR_CTE*), scaled by the amount of restated foreign earnings (*FORPT_EM*). See Figure 1 for complete variable definitions.

¹⁶ Although utilizing restatements of current tax expense to measure changes in taxable income mitigates much of the measurement error inherent in measuring taxable income, measurement error attributable to state and foreign taxes still exists.

¹⁷ Prior research has used deferred tax expense as a proxy for nonconforming earnings management (e.g., Phillips et al. 2003; Hanlon 2005). Deferred tax expense reflects earnings management that causes temporary book-tax differences, and thus excludes the possibility of earnings management that generates permanent book-tax differences. The valuation allowance account (VAA) induces additional measurement error when using deferred tax expense to estimate nonconforming earnings management, since changes in the VAA may be attributable to federal, foreign, or state tax jurisdictions, but firms disclose such jurisdictional information infrequently. Nonetheless, in supplemental analyses we compute a temporary nonconforming earnings management rate based on deferred tax expense. Inferences from empirical tests based on this alternate definition of nonconforming earnings management are substantially the same as those based on *NC_RATE* equals $1 - C_RATE$.

To investigate which firm-specific characteristics impact the trade-off between the tax benefits and expected detection costs associated with nonconforming earnings management (i.e., to test H1 and H2), we estimate the following regression using our sample of misstatement firms:

$$\begin{aligned}
 NC_RATE_i = & \alpha_0 + \alpha_1 NOL_i + \alpha_2 HI_CASH_i + \alpha_3 BIG_AUDIT_i + \alpha_4 AF_i + \alpha_5 FRAUD_i \\
 & + \alpha_6 MTB_i + \alpha_7 REV_ONLY_i + \alpha_8 EXP_ONLY_i + \alpha_9 LAG_TA_i + \alpha_{10} MNC_i + \alpha_{11} BLOAT_i \\
 & + \alpha_{12} LAG_NDTL_i + \alpha_{13} SC_PT_EM_i + \alpha_{14} INV_MILLS_i + \alpha_j \sum_j INDUS_i + \varepsilon_i
 \end{aligned} \quad (6)$$

where year t is the year for which firm i restates its financial statements. If firm i restates multiple years' financial statements, then year t is the first misstatement year. (See Figure 1 for complete variable definitions.)

NC_RATE , the dependent variable in equation (6), is the proportion (or rate) of nonconforming earnings management. We focus on the proportion rather than the total amount of nonconforming earnings management because our hypotheses focus on the rate of nonconforming earnings management, i.e., the amount of firm i 's nonconforming earnings management *relative* to its total earnings management.

To test H1 and H2, we include proxies for the net present value of tax benefits and the net expected detection costs associated with nonconforming earnings management. The first proxy for the net tax benefits is the availability of NOL carryforwards (NOL), which would absorb at least part of the impact of upward conforming earnings management and thereby reduce the tax benefits associated with *nonconforming* earnings management. All else equal, we expect misstatement firms with NOLs available at the beginning of the misstatement year to engage in less nonconforming upward earnings management than firms without such carryforwards.

Our second proxy for the net tax benefits of nonconforming earnings management is based on firms' free cash flow. All else equal, firms with relatively high free cash flow have a

greater ability to pay the additional taxes caused by conforming upward earnings management; this suggests they would apply lower discount rates and thus derive lower tax benefits from nonconforming earnings management than firms with relatively low levels of free cash flow. We define *HI_CASH* as an indicator variable equal to 1 if the firm is in the top one-third of the free cash flow distribution for all misstatement firms and predict that firms with *HI_CASH* equal to 1 engage in less nonconforming earnings management.¹⁸

We identify three proxies for the net expected detection costs associated with nonconforming earnings management. We assume that greater monitoring increases the probability of detection associated with nonconforming vis-à-vis conforming earnings management, and predict that misstatement firms subject to more effective monitoring are more likely to manage earnings in a conforming manner to minimize the risk of detection. Consistent with Francis and Wilson (1988) and Hanlon and Krishnan (2005), we assume that high quality auditors are more effective monitors and thus utilize the information in large book-tax differences more effectively than low quality auditors. Hence, we predict that all else equal, firms with high quality external auditors are less likely to manage earnings in a nonconforming manner. We define *BIG_AUDIT* as an indicator variable equal to 1 if the firm used a Big 4/5/6 auditor during the misstatement year. Similarly, greater analyst following (*AF*) should provide more effective monitoring (relative to less analyst following). Thus, we predict that all else equal, firms with greater analyst following are also less likely to engage in nonconforming earnings management. *AF* equals 1 (0) if a firm has any (no) analyst following.¹⁹

¹⁸ We obtain similar results when we use a 75 percent cutoff to determine when *HI_CASH* is equal to 1.

¹⁹ Results are similar when we set *AF* equal to the number of analysts following a firm. In untabulated results we also include institutional investors as a proxy for more effective monitoring. The results are similar to those tabulated in this paper, but the institutional investor variable is not significant, perhaps due to a reduced sample size.

Our third proxy for expected detection costs recognizes that the potential costs associated with fraudulent earnings management are undoubtedly very large, and we thus expect that firms engaging in fraud have strong incentives to avoid detection.²⁰ Moreover, we find in untabulated results that the termination of a top executive officer often accompanies the most egregious misstatements, which includes cases of fraudulent reporting. Consequently, we predict firms that commit accounting fraud and/or replace top executive officers in connection with earnings misstatements are more likely to have managed earnings in a book-tax conforming manner to minimize net expected detection costs. We label this detection cost variable *FRAUD* and define it as an indicator variable equal to 1 if a misstatement firm *ex post* was alleged to have engaged in egregious earnings management, which we proxy for by the presence of allegations of accounting fraud and/or the replacement of the firm's CEO, CFO, or President around the restatement date. To identify such firms, we use LexisNexis and 10kwizard.com to conduct a key word search for variants of "fraud" or "fictitious," "CEO," "CFO," or "President," and "firing," "dismissal," "removal," "step down," or "termination" in all misstatement firms' press releases and financial statements.²¹

Consistent with H1 and H2, we expect the coefficients on our proxies for the net tax benefits and the higher expected detection costs associated with nonconforming earnings management, i.e., $\alpha_1 - \alpha_5$ in equation (6), to be negative, indicative of less nonconforming earnings management.

²⁰ The excess costs of fraudulent earnings management include potentially higher legal and regulatory costs (e.g., attorney and court costs, and fines paid to regulatory agencies), shareholder wealth costs (due to the stock market's loss of confidence in the financial and/or tax reporting of the firm), and management turnover costs (due to terminating existing management and locating new, talented individuals).

²¹ Note that our fraud variable is based on allegations of fraud rather than legal outcomes, since the latter almost certainly would not occur until several years following the misstatement. Moreover, if fraud is suspected but is expected to be difficult or costly to prove, a firm (or prosecutor) might decide to not pursue a fraud case. Nonetheless, the firm would be more likely to replace any executives involved.

Equation (6) also includes control variables. First, because growth firms often report losses, it is possible that our NOL variable proxies for growth rather than the availability of tax benefit carryforwards. To address this issue, we include the market-to-book (*MTB*) ratio to control for the effect of growth opportunities. Second, there are likely different costs and benefits associated with managing different accounts to increase income, and thus we include additional indicator variables for misstatements that involve only revenue recognition irregularities (*REV_ONLY*) or only expense recognition irregularities (*EXP_ONLY*).

Third, we include controls for a firm's opportunities to manage earnings in conforming and nonconforming ways (i.e., for its earnings management opportunity set) as follows. *LAG_TA* and *MNC* control, respectively, for firm size and the firm's status as a multinational corporation. Larger firms and multinational firms presumably have different and greater opportunities to manage earnings.^{22, 23} *BLOAT* and *LAG_NDTL* capture the extent to which firms are constrained in their ability to manage earnings based on prior activity. *BLOAT* proxies for prior accrual manipulation (Barton and Simko 2002) and reflects the extent that a firm's balance sheet is already "bloated" with accruals, while *LAG_NDTL* captures the extent that a firm's balance sheet already contains higher deferred tax liabilities related to prior nonconforming earnings management (Phillips et al. 2004).²⁴ *SC_PT_EM* controls for the

²² Firm size can proxy for political costs (Watts and Zimmerman 1986). Large firms may engage in more conforming upward earnings management and thus incur higher current taxes and report higher current effective tax rates to avoid additional regulatory scrutiny (e.g., Zimmerman 1983), or they may be less inclined to manage earnings upwards (Watts and Zimmerman 1996).

²³ The U.S. does not tax foreign source income until it is repatriated to the U.S. but the same income is recognized on firms' consolidated financial statements. Multinational corporations can thus manage earnings in foreign countries to boost their book income but avoid any current U.S. tax consequences. In addition, *MNC* controls for the greater opportunities multinational firms have to manage taxable income (Rego 2003), which could increase temporary or permanent book-tax differences for reasons other than earnings management in shareholder reports.

²⁴ We include *LAG_NDTL* as the book-tax "bloat" control variable because permanent differences are not recognized in the financial statements and thus we can only control for the extent of prior *temporary* nonconforming earnings.

relative magnitude of the earnings management. The greater the amount of total earnings management scaled by total assets, the more likely managers will exhaust the desired earnings management method (e.g., nonconforming) available to them and thus employ more of the less desired method (e.g., conforming). Finally, we include *INDUS* to control for fundamental differences that may exist in earnings management opportunities across different industries.

Self-Selection Correction

Investigating the choice between conforming versus nonconforming earnings management among misstatement firms is problematic because the misstatement firms were likely ‘caught with their hands in the cookie jar’ by the SEC, their external auditor, or a corporate insider. To the extent the probability of detection is greater when a firm engages in nonconforming earnings management, the misstatement sample may reflect a self-selection bias towards nonconforming earnings management. Accordingly, we control for self-selection using Heckman’s (1979) two-step estimation approach.²⁵

More specifically, we first estimate a selection model to predict restatements, equation (7) below, using misstatement firms and a control sample of non-misstatement firms. The dependent variable in the selection model is *RESTATE*, which equals 1 if a firm restates earnings, and 0 otherwise. We estimate equation (7) and extract the self-selection adjustment, which is referred to as the inverse Mills ratio (*INV_MILLS*), and then include it in equation (6), which is the second step of the estimation procedure.²⁶ Assuming that self-selection bias is

²⁵ Unlike, for example, the choice of LIFO or FIFO (Maddala 1991), when a firm decides whether or not to manage earnings, it is not directly choosing to restate or not restate earnings. Rather, the firm in effect decides whether or not to take on the additional risk that its earnings management will be detected, which may then result in a restatement. Hence, self-selection bias in our setting may not be a first-order effect.

²⁶ *INV_MILLS* equals the probability density function of the standard normal divided by the cumulative density function (e.g., Kennedy 1998) extracted from the estimation of equation (7). We thank Shivaram Rajgopal for programming assistance in estimating *INV_MILLS*.

significant in our sample, we expect *NC_RATE* to be positively related to the inverse Mills ratio; i.e., *INV_MILLS* should be an increasing function of the probability that a firm is detected managing earnings due to nonconforming methods and thus is required to restate its earnings.

The selection model we estimate is as follows:

$$\begin{aligned} RESTATE_i = & \beta_0 + \beta_1 BT_DIFF_i + \beta_2 MJAC_i + \beta_3 MBE_i + \beta_4 MTB_i + \beta_5 FIN_R_i + \beta_6 LEV_i + \\ & \beta_7 INV_INTCOV_i + \beta_8 EPS_G_i + \beta_9 STRING_i + \beta_{10} CFO_i + \beta_{11} \Delta CFO_i + \\ & \beta_{12} BIG_AUDIT_i + \beta_{13} AF_i + \beta_{14} MNC_i + \beta_{15} LAG_TA_i + \beta_{16} \Delta AR_i + \\ & \beta_{17} \Delta INV_i + \beta_{18} \Delta CASHSALES_i + \beta_{19} \Delta EARN_i + \\ & \beta_{20} \Delta BN \Delta EMPLOYEES_i + \beta_{21} OP_LEASES_i + \beta_j \sum_j INDUS_i + \varepsilon_i, \end{aligned} \quad (7)$$

See Figure 1 for complete variable definitions.

Our restatement prediction model builds on prior restatement research by Richardson et al. (2002), on recent fraud research by Dechow et al. (2007), and on our previous discussion concerning the probability of detection associated with nonconforming earnings management. The explanatory variables in equation (7) represent the following factors: (a) Current period earnings management activity, as captured by modified Jones-model abnormal accruals, *MJAC* (Dechow et al. 1995) and current period book-tax differences, *BT_DIFF* (e.g., Phillips et al. 2003; Hanlon 2005).²⁷ (b) Incentives, or lack thereof, to manage earnings upwards, proxied by security issuances (*FIN_R*), closeness to debt covenants (*LEV* and *INV_INTCOV*), meeting current earnings targets (*MBE*), a history of meeting or beating analyst forecasts (*STRING*) and prior years' earnings targets (*EPS_G*), growth firms (*MTB*), and current operating performance (*ACFO* and *CFO*). (c) Proxies for the probability of detection, *BIG_AUDIT* and *AF*. (d)

²⁷ We estimate an alternative version of the model in which we substitute scaled deferred tax expense (*DTE* defined as *Compustat* #50 divided by #6, total assets) (Phillips et al. 2003) for *BT_DIFF*, which allows us to consider temporary rather than total book-tax differences.

Additional control variables that Dechow et al. (2007) find significant in predicting fraud (beyond the variables we have already included) and for which we have the data, including additional performance variables ($\Delta CASHSALES$ and $\Delta EARN$) and accrual quality variables (ΔAR and ΔINV), and variables reflecting non-financial (i.e., abnormal change in employees, $ABN\Delta EMPLOYEES$) and off-balance sheet (i.e., existence of operating leases, OP_LEASES) factors. (e) And other possible predictors of earnings misstatements, including firm size (LAG_TA), multinational activity (MNC), and industry membership ($INDUS$).²⁸

IV. SAMPLE

We identify misstatement firms (i.e., firms that restated their financial statements in a subsequent year) from the October 2002 GAO report. The report includes 845 firms that made 919 public announcements of restatements during the period from January 1, 1997, to June 30, 2002.²⁹ The restatements corrected previous material misstatements of financial statement numbers caused by accounting irregularities. Such irregularities occur when firms violate GAAP and reflect aggressive accounting practices, material errors, misinterpretation of accounting rules, or the misuse of facts and fraud.

The 845 firms' restatements reflect misstatements of annual or quarterly data. Because we require original and restated annual income tax data, we restrict our analysis to restatements of annual financial statements, which eliminates 407 observations. We hand-collect the

²⁸ According to Woolridge (2003, 585-587), all variables in the outcome equation should be included in the selection model; however, under certain circumstances it is permissible to omit such variables. In our case, we cannot include variables that we hand-collect due to the large number of observations in the non-restatement sample. Nonetheless, we also conducted our analysis by including all outcome equation variables available on *Compustat* in the selection model, and the results are similar.

²⁹ The GAO database excludes "restatements resulting from mergers and acquisition, discontinued operations, stock splits, issuance of stock dividends, currency-related issues, changes in business segment definitions, changes due to transfers of management, changes made for presentation purposes, general accounting changes under generally accepted accounting principles (GAAP), litigation settlements, and arithmetic and general bookkeeping errors" (GAO 2002). Hennes et al. (2007) provide evidence that restatements during our sample period were more likely caused by aggressive accounting practices relative to more recent restatements (i.e., GAO 2006), which are more likely caused by errors.

following variables from both the originally reported and restated financial reports: total assets, federal and foreign pretax net income, federal and foreign current tax expense, federal and foreign deferred tax expense, current and prior year net deferred tax liabilities, current and prior year deferred tax assets related to net operating losses, current and prior year deferred tax asset valuation allowance amounts, total revenue, income before extraordinary items, net income, and cash flow from operations. We lose 224 firms having incomplete original or restated balance sheet, income statement, statement of cash flow, and/or income tax footnote data. The full misstatement sample consists of 214 misstatement firms (see Table 1, Panel A).

INSERT TABLE 1 HERE

Panel B of Table 1 describes the composition of the full misstatement sample with regard to the direction of earnings management. Of the 214 misstatement firms, 151 managed earnings upwards and we concentrate our analysis on these firms.³⁰ That is, for the year (or the first year, if a firm has more than one misstatement year) that firms misstated their earnings, the restated pretax earnings are always lower than the amounts originally reported. We analyze the 151 first misstatement years because subsequent years of multi-year restatements likely reflect a mix of upward earnings management and accrual reversals from prior years' earnings management, and focusing on the first year of a multi-year restatement minimizes the impact of such reversals.³¹

In Table 1, Panel C, we summarize the types of accounting irregularities acknowledged by misstatement firms. Because some misstatements reflect multiple accounting irregularities,

³⁰ Panel B of Table 1 indicates that 37 firms restated earnings upwards. Motivations for downward earnings management appear to differ substantially from upward earnings management, and we leave an analysis of downward earnings managers to future research.

³¹ Untabulated results indicate that approximately 52 percent of misstatement firms restated for only one year, while 36 percent restated two years and 13 percent restated three years. In a sensitivity analysis, we examine firms that *consistently* managed earnings upward, and thus *consistently* restated their earnings downward; i.e., we include firms that restated earnings for multiple years where those restatements decreased pretax income in all years. The results (untabulated) from using this consistently income-increasing earnings management sample indicate that our main inferences remain unchanged.

there are 168 reasons provided by the 151 income-increasing misstatement firms. The most common accounting irregularities in our sample involve (to at least some extent) revenue recognition (40 percent) or expense recognition (20 percent). Per the GAO report, revenue recognition misstatements include instances in which revenue was improperly recognized, questionable revenues were recognized, or that were mistakes or improprieties that led to misreporting revenue.³² Expense misstatements include instances of improperly recognizing expenses, improperly capitalizing expenditures, or errors in reported expenses.

V. RESULTS

Univariate Analysis

Table 2, Panel A, presents descriptive statistics for our sample of income-increasing misstatement firms for *SC_PT_EM* (i.e., total pretax earnings restatement scaled by lagged total assets), *NC_RATE* and *C_RATE* (i.e., proportions, respectively, of nonconforming and conforming earnings management), variables that we predict affect the choice between the two earnings management types, and control variables. The mean (median) amount of *SC_PT_EM* for firms that managed earnings upward is 6.8 (2.5) percent of total assets.³³ The mean (median) *NC_RATE* of 62.9 (95.5) percent of total pretax earnings management is significantly greater than the mean (median) *C_RATE* of 37.1 (4.5) percent.³⁴ Figure 2 presents a histogram of the distribution of *NC_RATE* (and its complement *C_RATE*). Consistent with the median value of 95.5 percent, the majority of *NC_RATE* observations occur at or near a proportion of 1 (i.e., 100

³² Our sample contains 9 firms that restated earnings after SEC Staff Accounting Bulletin No. 101, *Revenue Recognition* (SEC 1999) was issued. Our basic inferences are unaffected when these firms are removed.

³³ Amounts of income-increasing earnings management are shown as positive amounts throughout.

³⁴ In untabulated analysis, we find that the vast majority of *NC_RATE* is attributable to temporary rather than permanent book-tax differences (respective means of 59 versus 4 percent, and medians of 92 versus 2 percent).

percent), but the histogram also displays a range of proportions, consistent with firms employing a mix of earnings management methods.³⁵

INSERT TABLE 2 HERE

INSERT FIGURE 2 HERE

Table 2, Panel A, also indicates that 53.1 percent of sample firms had NOL carryforwards available at the beginning of the misstatement year, the mean and median values of *FREECASH* were negative,³⁶ 85.3 percent of the sample firms used a Big 4/5/6 auditor, analysts followed 74.1 percent of the firms, and 24 percent of the misstatements reflected fraud and/or were associated with the firing of a top executive. Panel B shows that *NC_RATE* is significantly and negatively correlated (both Pearson and Spearman correlations) with the presence of NOLs and with high operating cash flows, the two factors we predict reduce the expected tax benefits related to nonconforming earnings management. *NC_RATE* is also significantly negatively correlated with the three factors that we predict increase expected detection costs (*BIG_AUDIT*, *FRAUD*, and *AF*), although for *AF* only the Spearman correlation is significant.³⁷

We provide additional univariate evidence by comparing the mean and median proportions of conforming and nonconforming earnings management (i.e., *C_RATE* and *NC_RATE*) in several contexts. Table 3, Panel A, displays the proportions of conforming and

³⁵ The histogram shows that *C_RATE* and *NC_RATE* are sometimes less than 0 or higher than 1. There are at least two reasons why such rates could occur: (1) restatements of accounting irregularities that have partially offsetting current and deferred tax effects, and (2) measurement error associated with our allocation of deferred tax assets related to NOL carryforwards (*ΔNOL*) between the federal and foreign jurisdictions. In the latter case, the measurement error could be caused by state tax complications or because the firm combined its NOL and other tax benefits (e.g., tax credit carryforwards) into a single deferred tax asset number.

³⁶ Note that by construction, one-third of the misstatement firms had high free cash flows.

³⁷ Untabulated correlations among the equation (6) independent variables indicate that virtually all correlations are small in magnitude. One exception is the Spearman correlation between *BLOAT* and *LAG_TA*, which is 0.609 (Pearson correlation is 0.345).

nonconforming earnings management by reason for the misstatement.³⁸ For the eight income-increasing misstatements that were due solely to irregularities in revenue recognition, all employed nonconforming earnings management strategies. Among the 18 firms having misstatements due solely to expense recognition irregularities, the mean and median *NC_RATEs* (73.4 and 96.7 percent, respectively) are significantly greater than the mean and median *C_RATEs* (26.6 and 0 percent). Moreover, this pattern of differences also holds for the remaining 125 misstatements that impact a variety of accounts.³⁹

Panel B of Table 3 presents the mean proportions of *NC_RATE* and *C_RATE* by industry for the misstatement firms. For the eight industries with at least five sample firms, *NC_RATE* exceeds *C_RATE* in all eight cases. Note that there is a mix of conforming and nonconforming earnings management across all industries represented in the sample.

Finally, Panel C of Table 3 shows the mean proportions of each earnings management type for the two tax benefit and three detection cost variables we predict reduce a firm's reliance on nonconforming earnings management. We expect that in the presence of each of these factors (e.g., *NOL* = 1), the mean *C_RATE* (*NC_RATE*) will be higher (lower) than when these factors are not present. The results are consistent with these expectations in eight of the ten comparisons; i.e., the proportion of conforming (nonconforming) earnings management increases (decreases) as each of the four factors (*NOL*, *HI_CASH*, *BIG_AUDIT*, and *FRAUD*) goes from a value of 0 to 1, and the differences are significant. The *AF* results are not significant.

INSERT TABLE 3 HERE

³⁸ Scholz (2008, 14) reports: "Ascertaining the accounting issues underlying these restatements can be difficult. Some companies describe errors by account (e.g., revenue is overstated), and others by the nature of the error (e.g., improper accounting for employee stock options). It is often unclear how the latter descriptions tie to specific financial statement elements, and thus it is not possible to consistently code by specific account."

³⁹ We obtain similar results when we use the rate of temporary nonconforming earnings management.

Multivariate Analysis

Table 4, Panels A and B, presents descriptive statistics for our misstatement firms (N = 151) and the control sample of non-misstatement firms (N = 7,948), respectively. The latter sample includes all *Compustat* firms not in the GAO (2002) restatement sample that are matched with the misstatement firms on industry (three-digit SIC) and fiscal year and that have the necessary data. Misstatement firms have significantly greater mean and median deferred tax expense (*DTE*), total book-tax differences (*BT_DIFF*), and abnormal accruals (*MJAC*) than non-misstatement firms. There is also evidence that misstatement firms have higher market-to-book ratios (*MTB*), a greater propensity to exceed the prior quarter's earnings (*EPS_G*) and beat analyst forecasts (*MBE*), a greater likelihood of raising funds externally (*FIN_R*) and of using operating leases (*OP_LEASES*), and greater increases in accounts receivable (ΔAR) and inventories (ΔINV) in year t . They also are less likely to use a Big 4/5/6 auditor (*BIG_AUDIT*) and be a multinational entity (*MNC*). On the other hand, misstatement firms are similar to the non-misstatement control firms on several important dimensions, including current tax expense, leverage, interest coverage, consecutive strings of meeting analyst forecasts, analyst following, firm size, cash flows from operations, and changes in CFO, earnings and number of employees. (Table 3, Panel B displays the industry composition for the misstatement and control samples.)

INSERT TABLE 4 HERE

Table 5, Panel A presents the results of estimating equation (7), our prediction model and step one in the Heckman (1979) two-step estimation approach. With regard to indicators of current period earnings management, the results suggest that coefficients on both *BT_DIFF* and *MJAC* are positive and significant, indicating that the book-tax difference variable is incrementally useful beyond abnormal accruals in predicting earnings restatements (and thus

income-increasing earnings management), and that *MJAC* is similarly incrementally useful to *BT_DIFF*. This result is consistent with total book-tax differences being useful in detecting upward earnings management.⁴⁰

INSERT TABLE 5 HERE

The results from estimating the prediction model in Panel A dealing with incentives (or the lack thereof) to manage earnings upward suggest the following significant effects: the probability of restatement is positively related to meeting or beating analyst forecasts (*MBE*) and prior quarterly earnings targets (*EPS_G*), and to seeking external financing (*FIN_R*). In addition, *RESTATE* is negatively related to the following variables: *BIG_AUDIT*, consistent with firms having high quality external auditors being less likely to restate earnings, and firm size (*LAG_TA*) and multinational activities (*MNC*), suggesting that larger firms and multinational firms are also less likely to restate. There is also evidence that *RESTATE* is more likely when receivables and inventories increase and operating cash flows decrease (ΔCFO). The last result suggests earnings management is less likely to occur when operating performance improves.⁴¹

Our main results reflect tests of H1 and H2 and are shown in Panel B of Table 5. These results are from estimating our multivariate earnings management strategy choice model, equation (6), which is the second step in the Heckman (1979) two-step approach.⁴² As predicted

⁴⁰ In untabulated results, we substitute deferred tax expense (*DTE*) for *BT_DIFF* in equation (7) to focus on temporary rather than total book-tax differences. The results yield similar inferences to those based on total book-tax differences (i.e., the coefficient on *DTE* is positive and significant). These results extend those in Phillips et al. (2003), which finds that deferred tax expense is incrementally useful to accrual-based variables in detecting upward earnings management to achieve certain earnings targets.

⁴¹ The results also show that restatements are more likely when earnings increase. We include both ΔCFO and $\Delta EARN$ in the prediction model; this implies that the coefficient on $\Delta EARN$ captures the change in accruals, and thus the results suggest that restatements are more likely when accruals increase, which is what we would expect. Note that the correlation between ΔCFO and $\Delta EARN$ is 0.42

⁴² The results in the text are based on estimating equation (6) using the inverse Mills ratio derived from estimating equation (7), which includes *BT_DIFF* as a regressor. Untabulated results from estimating equation (6) using the inverse Mills ratio derived from estimating equation (7) and including *DTE* instead of *BT_DIFF* are very similar.

by H1, the proportion of nonconforming earnings management declines in the presence of NOL carryforwards and high levels of free cash flow, although the results are significant at conventional levels only for NOL carryforwards. The coefficient on *NOL*, -0.192, suggests that firms with NOLs engage in 19.2 percent more book-tax *conforming* earnings management, which is consistent with firms engaging in less nonconforming earnings management when the net present value of the associated tax benefits is lower.⁴³

As predicted by H2, *NC_RATE* is lower when firms have a Big 4/5/6 auditor and when it is alleged that firms have committed fraud and/or a top executive is fired. The coefficient on *BIG_AUDIT* indicates that firms using a Big 4/5/6 auditor engage in 27.3 percent more book-tax conforming earnings management. Similarly, firms engaging in fraud or that fire a top executive in connection with the restatement employ 13.1 percent more conforming earnings management. The latter result is consistent with Erickson et al. (2004), who document that their sample of fraudulently reporting firms relied on substantial amounts of conforming earnings management. Overall, these results suggest that firms substitute conforming for nonconforming earnings management when net expected detection costs associated with nonconforming earnings

⁴³ In additional analyses, we redefine *NOL* by grossing-up deferred tax assets related to NOL carryforwards by the tax rate (35 percent) and then compute *NOL_Ratio*, which relates the estimated amount of NOL carryforwards to the amount of originally reported pretax income (which includes the amount of the upward managed earnings). For misstatement firms with positive pretax income, *NOL_Ratio* = 3.68 for firms that predominantly used conforming upwards earnings management versus *NOL_Ratio* = 0.75 for nonconforming misstatement firms, which is consistent with firms with sufficient amounts of NOL carryforwards relying on relatively more conforming earnings management since conforming earnings management imposes no additional tax costs on them. We then define *NOLRatio_Dum* to equal 1 if a firm's NOL carryforwards exceed the amount of original pretax income (and 0 otherwise). This variable indicates whether the firm has sufficient NOL carryforwards to absorb the effects of its upward conforming earnings management. We replace our *NOL* variable with *NOLRatio_Dum* and re-estimate equations (6) and (7). The results are consistent with those in Table 5, Panel B. In particular, the coefficients on *NOLRatio_Dum*, *BIG_AUDIT*, and *FRAUD* are all negative and significant at conventional levels.

management are higher, as they would be with high quality audit firms and for firms identified *ex post* as having engaged in the most egregious forms of earnings management.⁴⁴

Several control variables are also significant. First, after controlling for the kind of account misstated, the coefficient on *REV_ONLY* is significantly positive, contrary to conventional wisdom that revenue-related earnings management is mostly conforming. In addition, the coefficient on *EXP_ONLY* is also positive and significant. More importantly, these results indicate that even though a firm's propensity to engage in nonconforming earnings management strategies is in part determined by the kind of account managed, the effects predicted by H1 and H2 remain significant. We also find that *NC_RATE* is higher for firms with smaller *LAG_NDTL*, consistent with firms being less constrained to employ nonconforming earnings management if their balance sheet reflects less "bloat" with regard to deferred taxes. *NC_RATE* also increases with firm size, indicating that larger firms have more opportunities for nonconforming earnings management. Lastly, *INV_MILLS* is positive but insignificant, which suggests that self-selection bias is not a major problem in our sample, our prediction model is not very good, or the sample of misstatement firms is too small to meaningfully test *INV_MILLS*.⁴⁵

The multivariate results generally support H1 and H2 and suggest that firm-specific characteristics that reduce the net present value of tax benefits or increase the net expected

⁴⁴ We perform several untabulated analyses that yield qualitatively similar results. First, we examine *FRAUD* = 0 and *FRAUD* = 1 firms separately, and find that *NOL* and *BIG_AUDIT* are significant for the non-fraud and fraud firms, and *HI_CASH* is significant for fraud firms. Second, we re-run analyses after dropping the 13 firms with restatements reflecting 100 percent permanently nonconforming strategies. Third, we estimate the second-stage *NC_RATE* regression using logit, where the continuous *NC_RATE* variable is converted to 0 for values less than 50 percent (1 otherwise). Fourth, due to concerns regarding measurement error in computing *C_RATE* for firms with foreign operations, we re-run our main analyses excluding firms reporting foreign income (or loss).

⁴⁵ Stolzenberg and Relles (1997) argue that if selection bias is moderate or if samples have only a few hundred observations, which is the case here, then the two-step estimation approach can make estimates worse. In untabulated results we re-estimate equation (7) after excluding *INV_MILLS* and our main inferences are unchanged.

detection costs related to nonconforming earnings management are associated with firms that engage in more conforming and less nonconforming pretax earnings management.

VI. SUMMARY AND CONCLUSIONS

We examine tax and nontax implications of pretax earnings management by investigating the relative amounts of two types of earnings management: book-tax conforming and book-tax nonconforming. We base our analyses on a sample of firms that restated their earnings downward (and thus had managed earnings upwards), and we measure earnings management as the difference between originally reported and restated pretax earnings.

Using hand-collected data from the misstatement firms' financial reports, including originally reported and restated pretax income and current income tax expense, we calculate the relative amounts of both types of earnings management strategies. We find that nonconforming earnings management accounts for the larger proportion of total pretax earnings management, and we show that book-tax differences are incrementally useful in predicting earnings restatements. Most important, we provide evidence that firm-specific characteristics – presence of NOL carryforwards, use of a high quality external auditor, and more egregious earnings management – are associated with firms engaging in more conforming earnings management than they otherwise would have. Our results are consistent with misstatement firms trading off the net present value of tax benefits against the net expected detection costs associated with nonconforming earnings management when deciding whether to engage in conforming versus nonconforming upward earnings management strategies.

To our knowledge, this study is the first to investigate both tax *and* nontax factors that affect the conforming versus nonconforming choices that managers face when managing pretax earnings, and to utilize the restatement setting to more accurately measure both pretax earnings

management and changes in taxable income resulting from such earnings management. Our results lead to a better understanding of the book-tax trade-offs managers face when managing earnings upward. We bridge and contribute to two streams of literature: studies that document conforming earnings management to achieve tax savings, and studies where managers prefer to manage earnings in ways that do not incur current tax costs. By investigating the factors that lead to less nonconforming earnings management and utilizing a more general sample of earnings managers (misstatement firms in general versus firms committing fraud), we extend the work of Erickson et al. (2004).

We acknowledge that GAO misstatement firms may reflect some earnings management behavior not typically seen in the larger population of publicly-traded U.S. companies. Thus, our results may not generalize to the entire population. We note, however, that the most common kinds of accounting irregularities for our misstatement sample – revenue and expense recognition – are consistent with the most common areas of earnings management identified in Nelson et al. (2002) and Phillips et al. (2004). Finally, our analyses focus on pretax earnings management and thus we cannot speak to the relative magnitudes of pretax earnings management compared to income tax accrual management, as investigated by Dhaliwal et al. (2004), among others.

REFERENCES

- Badertscher, B., D. Collins, and T. Lys. 2007. Earnings management and the predictive ability of accruals with respect to future cash flows. Working Paper, University of Notre Dame, University of Iowa, and Northwestern University.
- Barth, M., D. Cram, and K. Nelson. 2001. Accruals and the prediction of future cash flows. *The Accounting Review* 76: 27-58.
- Barton, J., and P. Simko. 2002. The balance sheet as an earnings management constraint. *The Accounting Review* 77 (Supplement): 1-27.
- Becker, C., M. DeFond, J. Jiambalvo, and K. Subramanyam. 1998. The effect of audit quality on earnings management. *Contemporary Accounting Research* 15: 1-24.
- Burgstahler, D., and I. Dichev. 1997. Earnings management to avoid earnings decreases and losses. *Journal of Accounting and Economics* 24 (1): 99-128.
- Collins, D., and H. Wan, 2005. An empirical assessment of alternative discretionary accrual models. Working Paper, University of Iowa.
- DeAngelo, H., L. DeAngelo, and D. Skinner. 1994. Accounting choice in troubled companies. *Journal of Accounting & Economics* 17: 113-144.
- Dechow, P., W. Ge, C. Larson, and R. Sloan. 2007. Predicting Material Accounting Manipulations. Working paper, University of California, Berkley, University of Washington, University of Michigan, and Barclays Global Investors.
- Dechow, P., R. Sloan, and A. Sweeney. 1995. Detecting earnings management. *The Accounting Review* 70 (2): 193-225.
- DeFond, M., and J. Jiambalvo. 1994. Debt covenant violation and manipulation of accruals. *Journal of Accounting and Economics* 17: 145-177.
- Dhaliwal, D., M. Frankel, and R. Trezevant. 1994. The taxable and book income motivations for a LIFO layer liquidation. *Journal of Accounting Research* (2): 278-89.
- Dhaliwal, D., C. Gleason, and L. Mills. 2004. Last chance earnings management: using the tax expense to meet analysts' forecasts. *Contemporary Accounting Research* 21 (2): 431-457.
- Dhaliwal, D., R. Huber, H-S Lee, and M. Pincus. 2008. Variability of Book-Tax Differences, Information Uncertainty, and Implied Cost of Capital. Working paper, University of Arizona and University of California, Irvine.
- Durtschi, C., and P. Easton. 2005. Earnings management? The shapes of the frequency distributions of earnings metrics are not evidence ipso facto. *Journal of Accounting Research* 43 (4): 557-592.

Erickson, M., M. Hanlon, and E. Maydew. 2004. How much will firms pay for earnings that do not exist? Evidence of taxes paid on allegedly fraudulent earnings. *The Accounting Review* 79 (2): 387-408.

Francis, J., and E. Wilson. 1988. Auditor changes: A joint test of theories relating to agency costs and auditor differentiation. *The Accounting Review* 63 (4): 663-682.

Frank, M., and S. Rego. 2006. Do managers use the valuation allowance account to manage earnings around certain earnings targets? *Journal of the American Taxation Association* (forthcoming).

Frank, M., L. Lynch, and S. Rego. 2008. Does Aggressive Financial Reporting Accompany Aggressive Tax Reporting (and Vice Versa)? Working paper, University of Virginia and Iowa.

General Accounting Office. 2002. Financial statement restatements: Trends, market impacts, regulatory responses, and remaining challenges. Washington, DC: GAO-38-138.

General Accounting Office. 2006. Financial Statement Restatements: Updates of Public Company Trends, Market Impacts, and Regulatory Enforcement Actions. Washington, D. C.: GAO-06-678.

Guay, W., S.P. Kothari, and R. Watts. 1996. A market-based evaluation of discretionary accrual models. *Journal of Accounting Research* 34 (Supplement): 83-105.

Hanlon, M. 2003. What can we infer about a firm's taxable income from its financial statements? *National Tax Journal* 56 (December): 831-864.

Hanlon, M. 2005. The persistence and pricing of earnings, accruals, and cash flows when firms have large book-tax differences. *The Accounting Review* Vol. 80 (1).

Hanlon, M., and G. Krishnan. 2005. Do auditors use the information reflected in book-tax differences? Working paper, University of Michigan.

Hanlon, M., and E. Maydew, and T. Shevlin. 2006. Book-tax conformity and the information content of earnings. Working Paper, University of Michigan, University of North Carolina, and University of Washington.

Healy, P., and J. Wahlen. 1999. A review of the earnings management literature and its implications for standard setting. *Accounting Horizons* 13 (4): 365-383.

Heckman, J. 1979. Sample selection bias as a specification error. *Econometrica* 46 (6): 1251-1271.

Hennes, K., A. Leone, and B. Miller. 2007. The Importance of Distinguishing Errors from Irregularities in Restatement Research: The Case of Restatements and CEO/CFO Turnover. Working paper, Penn State University.

Johnson, W. B., and D. Dhaliwal. 1988. LIFO abandonment. *Journal of Accounting Research* 26 (2): 236-272.

Joos, P., J. Pratt, and S.D. Young. 2003. Using deferred taxes to infer the quality of accruals. Working paper, Massachusetts Institute of Technology.

Kennedy, P. 1998. *A Guide to Econometrics*. 4th edition. Cambridge, MA: MIT Press.

Krull, L. 2004. Permanently reinvested foreign earnings, taxes, and earnings management. *The Accounting Review* 79 (3): 745-768.

Maddala, G. 1991. A perspective on the use of limited-dependent and qualitative variables models in accounting research. *The Accounting Review* 66 (4): 788-807.

Matsunaga, S., T. Shevlin, and D. Shores. 1992. Disqualifying dispositions of incentive stock options: Tax benefits versus financial reporting costs. *Journal of Accounting Research* 30 (Supplement): 37-68.

Mills, L. 1998. Book-tax differences and Internal Revenue service adjustments. *Journal of Accounting Research* 36 (2): 343-356.

Mills, L., and K. Newberry. 2001. The influence of tax and nontax costs on book-tax reporting differences: Public and private firms. *Journal of the American Taxation Association* 23 (1): 1-19.

Mills, L., and R. Sansing. 2000. Strategic tax and financial reporting decisions: Theory and evidence. *Contemporary Accounting Research* 17(1): 85-106.

Nelson, M., J. Elliott and R. Tarpley. 2002. Evidence from auditors about managers' and auditors' earnings-management decisions. *The Accounting Review* 77 (Supplement): 175-202.

Palepu, Krishna G., Paul M. Healy, and Victor L. Bernard. 2003. *Business Analysis and Valuation Using Financial Statements: Text and Cases*. 3rd ed. Mason, Ohio: ThomsonSouthwestern.

Palmrose, Z.-V., V.J. Richardson, and S. Scholz. 2004. Determinants of market reactions to restatement announcements. *Journal of Accounting and Economics* 37 (1): 59-89.

Phillips, J., M. Pincus, and S. Rego. 2003. Earnings management: New evidence based on deferred tax expense. *The Accounting Review* 78 (2): 491-521.

Phillips, J., M. Pincus, S. Rego, and H. Wan. 2004. Decomposing changes in deferred tax assets and liabilities to isolate earnings management activities. *The Journal of the American Taxation Association* 26 (Supplement): 43-66.

- Rego, S. 2003. Tax-avoidance activities of U.S. multinational corporations. *Contemporary Accounting Research* 20 (4): 805-833.
- Revsine, L., D. Collins, and W. B. Johnson. 2005. *Financial Reporting and Analysis*. 3rd ed. Pearson Prentice-Hall.
- Richardson, S., I. Tuna, and M. Wu. 2002. Predicting earnings management: The case of earnings restatements. Working Paper, University of Pennsylvania and Hong Kong University of Science and Technology.
- Scholes, M., P. Wilson, and M. Wolfson. 1990. Tax planning, regulatory capital planning, and financial reporting strategy for commercial banks. *The Review of Financial Studies* 3 (4): 625-650.
- Scholz, S. 2008. The changing nature and consequences of public company restatements: 1997-2006. The Department of the Treasury (April).
- Securities and Exchange Commission (SEC). 1999. *Revenue Recognition*. Staff Accounting Bulletin No. 101. Washington, D.C.: Government Printing Office.
- Stolzenberg, R., and D. Relles. 1997. Tools for intuition about sample selection bias and its correction. *American Sociological Review* 62 (June): 494-507.
- Watts, R. and J. Zimmerman. 1986. *Positive Accounting Theory*. Prentice-Hall.
- Woolridge, J. 2003. *Introductory Econometrics: A Modern Approach*. 2nd ed. Mason, Ohio: ThomsonSouthwestern.
- Zimmerman, J. 1983. Taxes and firm size. *Journal of Accounting and Economics* 5 (2): 119-149.

TABLE 1
Misstatement Sample Selection and Description

Panel A: Misstatement Sample Selection	N
All GAO (2002) Restatement Announcements	845
Less:	
Exclusion of quarterly restatements	(407)
Exclusion of firms for insufficient financial, tax footnote, and/or restatement data	(224)
Misstatement Sample – Firms	214

Panel B: Misstatement Firms Partitioned by First Year Direction of Earnings Management	Firms
Direction of Earnings Management	
First-Year Income Increasing	151
First-Year Income Decreasing	37
No Effect on Pretax Income	26
Total	214

Panel C: Types of Accounting Irregularities Causing the Income-Increasing Misstatements	GAO Report	Percent of Obs.	Our Sample	Percent of Obs.
Reason for Misstatement				
Revenue only or revenue and other irregularities	388	36.09%	67	39.88%
Expense only or expense and other irregularities	208	15.70%	33	19.64%
Mergers and Acquisitions	64	5.93%	13	7.74%
In-Process Research and Development	36	3.57%	0	0.00%
Reclassification	48	5.27%	9	5.36%
Related Third Party	31	3.01%	5	2.98%
Restructuring	124	8.94%	16	9.52%
Securities	65	5.36%	10	5.95%
Other	111	14.21%	15	8.93%
Total Firm Irregularities	1075		168	100%
Total Firms	845		151	

TABLE 2
Descriptive Statistics for Income-Increasing Misstatement Firms (N = 151)

Panel A: Means, Medians, Percentiles, and Standard Deviations

Variable	Mean	Std. Dev.	75th	Percentiles Median	25th
<i>SC_PT_EM</i>	0.068	0.11	0.068	0.025	0.005
<i>NC_RATE</i>	62.86% ¹	0.56	100.00%	95.53% ¹	8.67%
<i>C_RATE</i>	37.14%	0.56	91.35%	4.46%	0.00%
<i>NOL</i>	0.531	0.501	1	1	0
<i>FREECASH</i>	-0.127	0.186	0.027	-0.058	-0.288
<i>HI_CASH</i>	0.333	--	--	--	--
<i>BIG_AUDIT</i>	0.853	0.355	1	1	0
<i>AF</i>	0.741	0.439	1	1	0
<i>FRAUD</i>	0.240	0.406	0	0	0
<i>MTB</i>	4.242	2.997	5.159	3.886**	2.048
<i>BLOAT</i>	1.656	0.684	1.465	0.749	0.371
<i>LAG_NDTL</i>	0.059	0.629	0.019	0.003	-0.003
<i>LAG_TA</i>	1,275	2,655	908	115	43
<i>MNC</i>	0.319	0.467	1	0	0

¹ A one-tail test of whether the mean (median) *NC_RATE* exceeds the mean (median) *C_RATE* is significant at the 0.01 level.

TABLE 2 (Continued)

Panel B: Pearson (above diagonal) and Spearman (below diagonal) Correlations (p-values shown below correlations) for Income-Increasing Misstatement Firms

	<i>NC_RATE</i>	<i>NOL</i>	<i>HI_CASH</i>	<i>BIG_AUD</i>	<i>AF</i>	<i>FRAUD</i>
<i>NC_RATE</i>	---	-0.202 0.001	-0.122 0.050	-0.192 0.003	-0.102 0.146	-0.165 0.047
<i>NOL</i>	-0.219 0.001	---	-0.143 0.089	0.125 0.136	-0.164 0.050	0.013 0.442
<i>HI_CASH</i>	-0.108 0.087	-0.143 0.089	---	0.051 0.355	0.134 0.076	-0.117 0.033
<i>BIG_AUDIT</i>	-0.196 0.001	0.125 0.136	0.051 0.355	---	0.232 0.005	-0.087 0.098
<i>AF</i>	-0.222 0.019	-0.164 0.050	0.134 0.076	0.232 0.005	---	0.037 0.316
<i>FRAUD</i>	-0.183 0.041	0.013 0.442	-0.117 0.033	-0.087 0.098	0.037 0.316	---

See Figure 1 for complete variable definitions.

TABLE 3
Proportions of Conforming and Nonconforming Earnings Management

Panel A: By Reason for Misstatement

	Observations	<i>C_RATE</i>		<i>NC_RATE</i>	
		Mean	Median	Mean	Median
Revenue Only	8	0.0%***	0.0%***	100.0%	100.0%
Expense Only	18	26.6%***	0.0%***	73.4%	96.7%
All other Combinations	125	39.1%***	15.2%***	60.9%	84.7%
Total Firms	151				

*, **, *** Statistically significant at the 0.10, 0.05, 0.01 level, respectively, using a one-tailed test of differences in proportions within each row. For example, *C_RATE* for Revenue Only (0.0%) is compared to *NC_RATE* for Revenue Only (100.0%).

Panel B: By Industry Classification

Industry	Non-Misstatement Sample ¹		Misstatement Sample			
	# Obs	Percent	# Obs	Percent	<i>C_RATE</i>	<i>NC_RATE</i>
Transportation	234	2.94%	5	3.31%	11.23%	88.77%
Utilities	189	2.38%	3	1.99%	96.70%	3.30%
Services	903	11.36%	28	18.54%	39.30%	60.70%
Retail	755	9.50%	24	15.89%	28.20%	71.80%
Food	5	0.06%	2	1.32%	72.20%	27.80%
Extractive	191	2.40%	2	1.32%	0.27%	99.73%
Textiles and Printing/Publishing	162	2.04%	7	4.64%	41.36%	58.64%
Chemicals	118	1.48%	2	1.32%	100.00%	0.00%
Computers	1,985	24.97%	29	19.21%	33.10%	66.90%
Pharmaceuticals	670	8.43%	5	3.31%	19.43%	80.57%
Mining and Construction	64	0.81%	4	2.65%	40.38%	59.62%
Durable Manufacturers	2,421	30.46%	32	21.19%	44.20%	55.80%
Financials	251	3.16%	8	5.30%	29.62%	70.38%
Total Observations	7,948	100%	151	100%		

¹ For descriptive purposes, we use Barth et al.'s (2001) industry classifications: Mining & Construction (1000-1999, excluding 1300-1399), Food (2000-2111), Textiles & Printing/Publishing (2200-2780), Chemicals (2800-2824, 2840-2899), Pharmaceuticals (2830-2836), Extractive (2900-2999), 1300-1399), Durable Manufactures (3000-3999, excluding 3570-3579 and 3670-3679), Computers (7370-7379, 3570-3579, 3670-3679), Transportation (4000-4899), Utilities (4900-4999), Retail (5000-5999), and Services (7000-8999), excluding 7370-7379).

TABLE 3 (Continued)

Panel C: By Partitioning Based on *NOL*, *HI_CASH*, *BIG_AUDIT*, *AF*, and *FRAUD*

Proxies for Net Present Value of Tax Benefits Related to Nonconforming Earnings Management			Proxies for Net Expected Detection Costs Related to Nonconforming Earnings Management		
	<i>C_RATE</i>	<i>NC_RATE</i>		<i>C_RATE</i>	<i>NC_RATE</i>
<i>NOL</i> = 0	36.2%***	63.8%***	<i>BIG_AUDIT</i> = 0	20.4%***	79.6%***
<i>NOL</i> = 1	47.9%	52.1%	<i>BIG_AUDIT</i> = 1	40.3%	59.7%
	<i>C_RATE</i>	<i>NC_RATE</i>		<i>C_RATE</i>	<i>NC_RATE</i>
<i>HI_CASH</i> = 0	42.3%*	57.7%*	<i>AF</i> = 0	33.1%	66.9%
<i>HI_CASH</i> = 1	51.8%	48.2%	<i>AF</i> = 1	38.6%	61.5%
				<i>C_RATE</i>	<i>NC_RATE</i>
			<i>FRAUD</i> = 0	35.0%***	65.0%***
			<i>FRAUD</i> = 1	48.7%	51.3%

*, **, *** Statistically significant at the 0.10, 0.05, 0.01 level, respectively, using a one-tailed test of differences in proportions for each column. For example, *C_RATE* for *NOL* = 0 (36.2%) is compared to *C_RATE* for *NOL* = 1 (47.9%).

See Figure 1 for complete variable definitions.

TABLE 4
Comparative Descriptive Statistics for Income-Increasing Misstatement Firms versus a Control Sample of Non-Misstatement Firms

Panel A: Misstatement Firms

	N	Mean	Std. Dev.	Percentiles		
				75 th	50 th	25 th
<i>CTE</i>	151	0.017	0.033	0.031	0.005	0.000
<i>DTE</i>	151	-0.003**	0.027	0.004	0.001*	-0.007
<i>BT_DIFF</i>	151	0.018***	0.173	0.082	0.004**	-0.077
<i>MJAC</i>	151	0.048***	0.232	0.096	0.031***	-0.064
<i>MBE</i>	151	0.697**	0.435	1	1	0
<i>MTB</i>	151	4.242**	2.997	5.159	3.886**	2.048
<i>FIN_R</i>	151	0.311***	0.346	0.417	0.115	0.029
<i>LEV</i>	151	0.25	0.211	0.391	0.221	0.068
<i>INV_INTCOV</i>	151	0.541	0.889	0.359	0.259	0.116
<i>EPS_G</i>	151	2.546**	1.875	4	2*	1
<i>STRING</i>	151	3.44	2.798	5	2	1
<i>CFO</i>	151	-0.0219	0.114	0.085	0.018	-0.061
<i>ΔCFO</i>	151	-0.049	0.167	0.033	-0.019	-0.133
<i>BIG_AUDIT</i>	151	0.748***	0.355	1	1	0
<i>AF</i>	151	0.682	0.515	1	1	0
<i>MNC</i>	151	0.319**	0.467	1	0	0
<i>LAG_TA</i>	151	1,275	2,655	908	115	43
<i>ΔAR</i>	151	0.059**	0.184	0.049	0.032***	0.013
<i>ΔINV</i>	151	0.037***	0.181	0.034	0.006**	0.001
<i>ΔCASHSALES</i>	151	0.242	0.196	0.201	0.143	0.034
<i>ΔEARN</i>	151	-0.016	0.205	0.012	-0.011	-0.007
<i>ABNΔEMPLOYEES</i>	151	-0.0283	0.318	0.005	-0.006	-0.016
<i>OP_LEASES</i>	151	0.888**	0.316	1	1	1

TABLE 4 (Continued)

Panel B: Non-Misstatement Firms^a

	N	Mean	Std. Dev.	75 th	Percentiles 50 th	25 th
<i>CTE</i>	7,948	0.022	0.023	0.024	0.011	0.000
<i>DTE</i>	7,948	-0.013	0.022	0.006	-0.021	-0.127
<i>BT_DIFF</i>	7,948	-0.011	0.168	0.052	-0.015	-0.096
<i>MJAC</i>	7,948	-0.065	0.243	0.125	-0.050	-0.289
<i>MBE</i>	7,948	0.513	0.422	1	1	0
<i>MTB</i>	7,948	3.646	4.814	4.163	2.023	1.033
<i>FIN_R</i>	7,948	0.213	0.217	0.289	0.067	0.044
<i>LEV</i>	7,948	0.293	0.254	0.542	0.292	0.046
<i>INV_INTCOV</i>	7,948	0.541	0.879	0.345	0.256	0.136
<i>EPS_G</i>	7,948	2.063	1.925	3	1	0
<i>STRING</i>	7,948	2.654	2.740	5	2	0
<i>CFO</i>	7,948	-0.031	0.223	0.071	0.017	-0.183
<i>ΔCFO</i>	7,948	0.028	0.214	0.033	0.000	-0.073
<i>BIG_AUDIT</i>	7,948	0.872	0.323	1	1	1
<i>AF</i>	7,948	0.781	0.471	1	1	1
<i>MNC</i>	7,948	0.485	0.499	1	0	0
<i>LAG_TA</i>	7,948	1,627	2,948	1,504	543	54
<i>ΔAR</i>	7,948	0.025	0.154	0.028	0.013	0.002
<i>ΔINV</i>	7,948	0.002	0.152	0.022	0.002	0.000
<i>ΔCASHSALES</i>	7,948	0.186	0.176	0.165	0.077	0.016
<i>ΔEARN</i>	7,948	-0.009	0.153	0.009	-0.007	-0.003
<i>ABNΔEMPLOYEES</i>	7,948	-0.018	0.259	0.103	-0.019	-0.109
<i>OP_LEASES</i>	7,948	0.763	0.187	1	1	1

^a Non-misstatement firms represent all *Compustat* firms not appearing in the GAO restatement sample and matched on three-digit SIC and year of the misstatement.

*, **, *** Statistically significant at the 0.10, 0.05, 0.01 level, respectively, between misstatement and non-misstatement firms.

See Figure 1 for complete variable definitions.

TABLE 5
Results from Heckman (1979) Two-Step Regressions

Panel A: Results from First Stage of Heckman Estimation – Selection Model

$$\begin{aligned}
 RESTATE_i = & \beta_0 + \beta_1 BT_DIFF_i + \beta_2 MJAC_i + \beta_3 MBE_i + \beta_4 MTB_i + \beta_5 FIN_R_i + \beta_6 LEV_i + \beta_7 INV_INTCOV_i \\
 & + \beta_8 EPS_G_i + \beta_9 STRING_i + \beta_{10} CFO_i + \beta_{11} \Delta CFO_i + \beta_{12} BIG_AUDIT_i + \beta_{13} AF_i + \beta_{14} MNC_i + \beta_{15} LAG_TA_i \\
 & + \beta_{16} \Delta AR_i + \beta_{17} \Delta INV_i + \beta_{18} \Delta CASHSALES_i + \beta_{19} \Delta EARN_i + \beta_{20} ABN\Delta EMPLOYEES_i + \beta_{21} OP_LEASES_i \\
 & + \beta_j \Sigma_j INDUS_i + \varepsilon_i, \quad (7)
 \end{aligned}$$

	Predicted Sign	Coefficient	p-value
<i>Intercept</i>	+/-	-1.942	0.001
<i>BT_DIFF</i>	+	2.781	0.042
<i>MJAC</i>	+	0.588	0.067
<i>MBE</i>	+	0.198	0.067
<i>MTB</i>	+	0.061	0.331
<i>FIN_R</i>	+	1.187	0.001
<i>LEV</i>	+	-0.020	0.485
<i>INV_INTCOV</i>	+	0.192	0.380
<i>EPS_G</i>	+	0.282	0.045
<i>STRING</i>	+	0.037	0.275
<i>CFO</i>	-	-0.005	0.462
<i>ΔCFO</i>	-	-1.417	0.077
<i>BIG_AUDIT</i>	-	-0.147	0.007
<i>AF</i>	-	0.031	0.119
<i>MNC</i>	+/-	-0.179	0.078
<i>LAG_TA</i>	+/-	-0.026	0.027
<i>ΔAR</i>	+	1.145	0.017
<i>ΔINV</i>	+	2.089	0.005
<i>ΔCASHSALES</i>	+	-0.023	0.388
<i>ΔEARN</i>	-	1.988	0.049
<i>ABNΔEMPLOYEES</i>	-	-0.122	0.445
<i>OP_LEASES</i>	+	0.268	0.352
Pseudo R ²		31%	

N = 151 misstatement firms and N = 7,948 non-misstatement firms.

Panel B: Results From Second Stage of Heckman Estimation – Choice Model

$$NC_RATE_i = \alpha_0 + \alpha_1 NOL_i + \alpha_2 HI_CASH_i + \alpha_3 BIG_AUDIT_i + \alpha_4 AF_i + \alpha_5 FRAUD_i + \alpha_6 MTB_i + \alpha_7 BLOAT_i + \alpha_8 LAG_NDTL_i + \alpha_9 LAG_TA_i + \alpha_{10} SC_PT_EM_i + \alpha_{11} MNC_i + \alpha_{12} REV_ONLY_i + \alpha_{13} EXP_ONLY_i + \alpha_{14} INV_MILLS_i + \alpha_j \Sigma_j INDUS_i + \varepsilon_i \quad (6)$$

	Predicted Sign	Coefficient	p-value
Intercept	+/-	0.549	0.013
<i>NOL</i>	-	-0.192	0.024
<i>HI_CASH</i>	-	-0.113	0.127
<i>BIG_AUDIT</i>	-	-0.273	0.037
<i>AF</i>	-	-0.022	0.402
<i>FRAUD</i>	-	-0.131	0.048
<i>MTB</i>	-	-0.003	0.214
<i>BLOAT</i>	-	0.000	0.318
<i>LAG_NDTL</i>	-	-0.091	0.054
<i>LAG_TA</i>	+/-	0.044	0.042
<i>SC_PT_EM</i>	-	0.193	0.260
<i>MNC</i>	+/-	-0.018	0.438
<i>REV_ONLY</i>	+/-	0.498	0.034
<i>EXP_ONLY</i>	+/-	0.121	0.092
<i>INV_MILLS</i>	+	0.014	0.323
Adjusted R ²		35%	
<i>N</i>		151	

Industry coefficients are included but not reported.

p-values are one-tail if there is a predicted sign, two-tail otherwise.

See Figure 1 for complete variable definitions.

APPENDIX A

Present Value of Tax Benefits Associated with Nonconforming Earnings Management

Assume that in the current year t and also in year $t + n$, a firm has \$100 of “true” pretax book income. The firm engages in \$10 of earnings management and thus originally reports \$110 of pretax income in year t . Assume further that the earnings management reverses in year $t + n$ and the firm reports pretax income of \$90.

If the above earnings management is performed in a conforming manner and the income tax rate is 35 percent, then the firm would incur tax costs of \$38.50 in year t ($\110×35 percent) and \$31.50 in year $t + n$ ($\$90 \times 35$ percent). The present value of tax costs would be $\$38.50 + [\$31.50 / (1+r)^n]$, where r is the firm’s discount rate.

If the earnings management is performed in a nonconforming manner, then the firm would incur tax costs of \$35 ($\$100 \times 35\%$) in both years t and $t + n$, since nonconforming earnings management does not affect taxable income (in this case, \$100). The present value of tax costs would be $\$35 + [\$35 / (1+r)^n]$. If the nonconforming earnings management results in a temporary book-tax difference, then the second term takes on a positive value; however, if the nonconformity results in a permanent book-tax difference, the second term drops out and the tax costs would be \$35.

The tax benefit from engaging in nonconforming versus conforming upward earnings management equals the net present value of tax benefits from doing so, i.e., the present value of tax costs associated with conforming earnings management minus the present value of tax costs associated with nonconforming earnings management. This amount is computed as:

$$\begin{aligned} & \{ \$38.50 + [\$31.50 / (1+r)^n] \} - \{ \$35 + [\$35 / (1+r)^n] \} \\ & = \$3.50 - [\$3.50 / (1+r)^n]. \end{aligned}$$

More formally, the present value of tax benefits from managing earnings in a nonconforming manner in the current period is:

$$PVTB_{NC} = PT_EM \times STR \{1 - [1 / (1+r)^n]\}$$

where PT_EM is the amount of pretax earnings management and STR is the maximum corporate statutory tax rate (35 percent). Note that:

$$\partial PVTB_{NC} / \partial r > 0;$$

$$\partial PVTB_{NC} / \partial n > 0;$$

$$\partial PVTB_{NC} / \partial PT_EM > 0; \text{ and}$$

$$\partial PVTB_{NC} / \partial STR > 0.$$

APPENDIX B

Detailed Computation of *CONFORM_EM*

Assume a firm has \$80 of “true” pretax book income and engages in \$10 each of conforming and nonconforming earnings management, and thus originally reports \$100 of pretax income. For simplicity, assume the firm has no temporary book-tax differences except those created by earnings management, has no permanent book-tax differences, and has no foreign earnings. Further, assume the firm has no deferred tax liabilities or deferred tax asset valuation allowance account (VAA) at the beginning of the year, and the firm’s only deferred tax asset of \$350 relates to a \$1,000 NOL carryforward. The applicable statutory tax rate is 35 percent.

When its accounting irregularities become known, the firm restates its pretax income to \$80. The firm originally reported \$90 of taxable income (\$80 of pretax income plus \$10 of conforming earnings management). The firm has no current tax expense given its NOL carryforwards, which were originally reduced to \$910 (from \$1,000) and valued at \$318.50 ($\$910 \times 35\%$) on the year-end balance sheet as a deferred tax asset. The \$10 of nonconforming earnings management created a \$3.50 ($\$10 \times 35\%$) deferred tax liability. These items lead to \$35.00 of originally reported deferred tax expense.

Upon restatement, the firm only has \$80 of taxable income because there are no book-tax differences. The firm still reports zero current tax expense, but the \$1,000 NOL carryforward is now only reduced to \$920 and valued at \$322 ($\$920 \times 35\%$) on the year-end balance sheet. In addition, the firm no longer has any deferred tax liability at year-end because the earnings restatement removed the temporary book-tax differences. These items lead to \$28 of restated deferred tax expense.

The originally reported and restated pretax income and income tax expense amounts are as follows:

	<u>Original</u>	<u>Restated</u>	<u>Difference</u>
Pretax Income	<u>\$100.00</u>	<u>\$80.00</u>	<u>\$20.00</u>
Current Tax Expense	0.00	0.00	0.00
Deferred Tax Expense	<u>35.00</u>	<u>28.00</u>	<u>7.00</u>
Total Tax Expense	<u>35.00</u>	<u>28.00</u>	<u>\$7.00</u>
Net Income Per Books	<u>\$65.00</u>	<u>\$52.00</u>	
Taxable Income	<u>\$90.00</u>	<u>\$80.00</u>	

The difference between originally reported and restated pretax income (PT_EM) is \$20; the difference between originally reported and restated federal current tax expense (ΔFED_CTE) is \$0.⁴⁶ We emphasize that the restatement of deferred tax assets related to NOL carryforwards erroneously causes the change in current tax expense to understate the tax effect of conforming earnings management (\$0 instead of \$3.50) and the change in deferred tax expense to overstate the tax effect of nonconforming earnings management (\$7 instead of \$3.50). We correct for such measurement error by computing our proxies for conforming and nonconforming earnings management. First, we compute $\Delta FEDNOL_DTA$ as the difference between the original and restated change in the NOL deferred tax asset or $[(\$318.50 - \$350.00) - (\$322.00 - \$350.00)]$, which equals -\$3.50.

With $\Delta FEDNOL_DTA$ equal to -\$3.50, we then compute the dollar amount of the conforming earnings management proxy and the rates of conforming and nonconforming earnings management as follows:

$$CONFORM_EM = (\Delta FED_CTE - \Delta FEDNOL_DTA) / FED_RATE$$

$$= [0 - (-\$3.50)] / .35 = \$3.50 / .35 = \$10;$$

$$C_RATE = \$10/\$20 = 0.5; \text{ and}$$

$$NC_RATE = 1 - 0.5 = 0.5.$$

⁴⁶ See Figure 1 for complete variable definitions.

APPENDIX C

Two Examples from our Misstatement Sample

Versatility Inc. restated its fiscal year 1996 financial statements for the accounting treatment of certain transactions including revenue recognition criteria and timing, bad debt reserves, and other expense-related items. We determine that approximately 73 percent of Versatility's earnings management was book-tax conforming, as shown below.

<i>Versatility Inc.</i>			
Year: 1996 (in millions)	Original	Restated	Difference
Federal Pretax Net Income	3.01	-8.87	11.88
Foreign Pretax Net Income	-0.10	0.70	-0.80
Total Pretax Net Income	2.91	-8.18	11.09
Current Tax Provision:			
Federal	1.02	-0.33	1.35
Foreign	0.01	0.1	-0.09
State	0.23	-0.1	0.33
Total Current Provision	1.26	-0.32	1.59
Deferred Tax Provision:			
Federal	-0.24	0	-0.24
Foreign	0	0	0
State	-0.06	0	-0.06
Total Deferred Provision	-0.3	0	-0.3
Total Tax Provision	0.96	-0.32	1.29
DTAs from NOLs at year-end <i>t</i>	0	1.76	-1.76
DTAs from NOLs at year-end <i>t-1</i>	0	0	0
Change in DTAs from NOLs	0	1.76	-1.76
<i>C_RATE</i>			73.05%⁴⁷
<i>NC_RATE</i>			26.95%

⁴⁷ To compute *C_RATE*, we estimate the foreign tax rate that applies to the restated foreign earnings as $FOR_RATE = \Delta FOR_CTE / \Delta FORPT_EM = -0.09 / -0.80 = 0.1125$. Because deferred tax assets related to foreign NOLs are zero at the beginning of the year, we allocate the total decrease in deferred tax assets related to NOLs to the U.S. jurisdiction (i.e., $\Delta FEDNOL_DTA = -1.76$ and $\Delta FORNOL_DTA = 0.00$). Using these components and data from the table above, we compute $C_RATE = \{[(\Delta FEDCTE - \Delta FEDNOL_DTA) / .35] + [(\Delta FORCTE - \Delta FORNOL_DTA) / FOR_RATE]\} / \Delta PPT_EM = \{[(1.35 - (-1.76)) / .35] + [(-0.09 - 0) / (0.1125)]\} / 11.09 = 73.05\%$.

Pre-Paid Legal Services Inc. restated its 1999 financial statements and significantly lowered reported earnings per share due to revenue and expense recognition practices. We determine that approximately 97 percent of Pre-Paid Legal Services' earnings management was book-tax nonconforming, as shown below.

<i>Pre-Paid Legal Services, Inc.</i>			
Year: 1999 (in millions)	Original	Restated	Difference
Federal Pretax Net Income	59.93	18.51	41.42
Foreign Pretax Net Income	0.0	0.0	0.0
Total Pretax Net Income	59.93	18.51	41.42
Current Tax Provision:			
Federal	5.34	4.9	0.44
State	1.07	1.07	0
Total Current Provision	6.41	5.96	0.44
Deferred Tax Provision:			
Federal	14.57	0.52	14.05
Total Deferred Provision	14.57	0.52	14.05
Total Tax Provision	20.97	6.48	14.49
DTAs from NOLs at year-end <i>t</i>	0	0	0
DTAs from NOLs at year-end <i>t-1</i>	1.05	1.05	0
Change in DTAs from NOLs	-1.05	-1.05	0
C_RATE			3.04% ⁴⁸
NC_RATE			96.96%

⁴⁸ To compute *C_RATE*, we estimate the foreign tax rate that applies to the restated foreign earnings as $FOR_RATE = \Delta FOR_CTE / \Delta FORPT_EM = 0$, based on the information above. Because there is no change in deferred tax assets related to NOLs, $\Delta FEDNOL_DTA = 0$ and $\Delta FORNOL_DTA = 0$. Using these components and data from the table above, we compute $C_RATE = \{[(\Delta FEDCTE - \Delta FEDNOL_DTA) / .35] + [(\Delta FORCTE - \Delta FORNOL_DTA) / FOR_RATE]\} / \Delta PPT_EM = \{(0.44 - 0) / .35 + 0\} / 41.42 = 3.04\%$.

FIGURE 1 Variable Definitions

Variables to Compute *CONFORM_EM* (Conforming Earnings Management):

$$CONFORM_EM = \frac{(\Delta FED_CTE - \Delta FEDNOL_DTA)}{FED_RATE} + \frac{(\Delta FOR_CTE - \Delta FORNOL_DTA)}{FOR_RATE}$$

ΔFED_CTE = Difference between the originally reported and restated federal current tax expense.

ΔFEDNOL_DTA = The change in federal deferred tax assets related to NOL carryforwards computed as $\Delta NOL_DTA \times (FEDPT_EM / PT_EM)$.

ΔNOL_DTA = Difference between the originally reported and restated change in deferred tax assets related to NOL carryforwards.

FEDPT_EM = Difference between the originally reported and restated U.S. pretax income.

PT_EM = Difference between the originally reported and restated total pretax income.

FED_RATE = The maximum corporate federal statutory tax rate, which was 35 percent throughout our sample period.

ΔFOR_CTE = Difference between the originally reported and restated foreign current tax expense.

ΔFORNOL_DTA = The change in foreign deferred tax assets related to NOL carryforwards computed as $\Delta NOL_DTA \times (FORPT_EM / PT_EM)$.

FORPT_EM = Difference between the originally reported and restated foreign pretax income.

FOR_RATE = Foreign tax rate computed as $(\Delta FOR_CTE) / (FORPT_EM)$.

Variables Used in Equation (6):

NC_RATE = Firm *i*'s proportion of nonconforming earnings management, computed as $(1 - C_RATE)$ in year *t*; $C_RATE = CONFORM_EM / PT_EM$.

NOL = 1 if firm *i* has net operating loss carryforwards available at the beginning of year *t*, and 0 otherwise.

HI_CASH = 1 if firm *i* is in the top one-third of the *FREECASH* distribution for misstatement firms during year *t-1*, and 0 otherwise. *FREECASH* = Cash flow from operating activities (*Compustat* #308) minus capital expenditures (#128) plus sales of property, plant, and equipment (#107) minus acquisitions (#129) in year *t-1*, scaled by total assets (#6) at year-end *t-2*.

BIG_AUDIT = 1 if firm *i* is audited by a Big 4/5/6 auditor (#149 value of 1 – 7) in year *t*, and 0 otherwise.

AF = 1 if firm *i* has any analyst following in year *t*, and zero otherwise.

FRAUD = 1 if a misstatement firm *ex post* was alleged to have engaged in accounting fraud and/or the misstatement firm replaced its CEO, CFO, or President in association with the restatement, and 0 otherwise.

MTB = Firm *i*'s market-to-book ratio, calculated as market value of equity (#25 × #199) at year-end *t-1* divided by the book value of equity (#60) at year-end *t-1*.

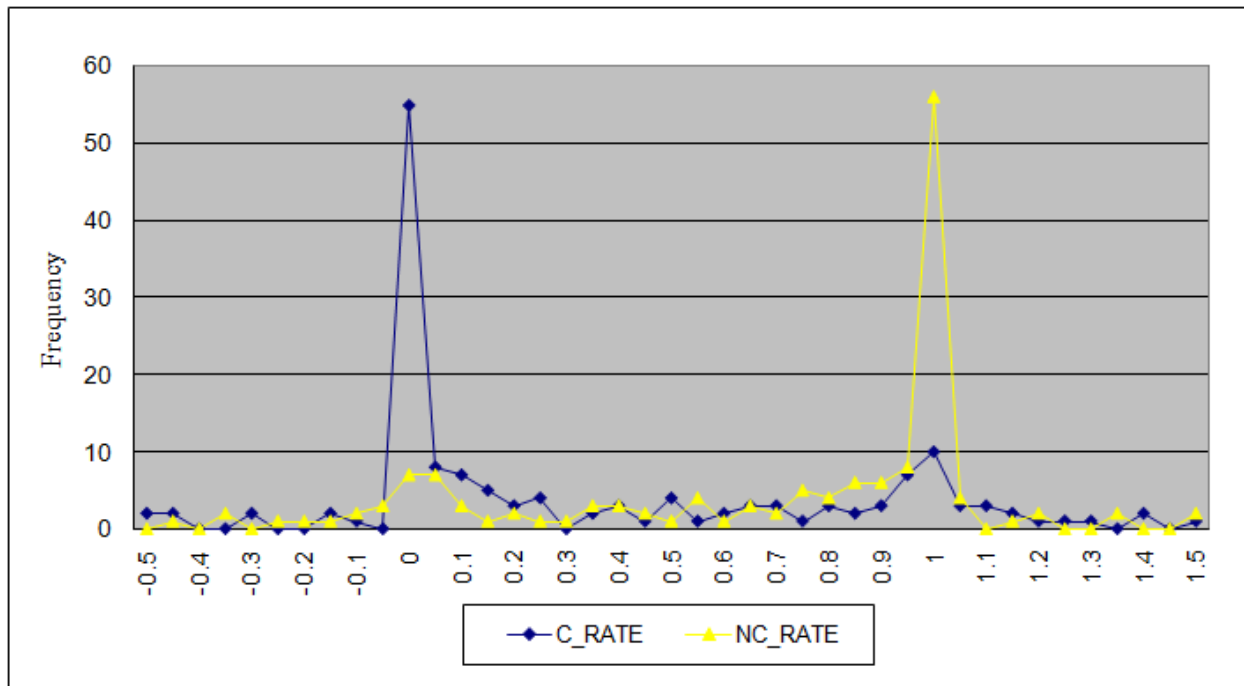
<i>REV_ONLY</i>	= 1 if the difference between the original and restated pretax net income is equal to the difference between the original and restated revenue for firm <i>i</i> in year <i>t</i> , and 0 otherwise.
<i>EXP_ONLY</i>	= 1 if the GAO report indicates that firm <i>i</i> only restated expense accounts in year <i>t</i> , and 0 otherwise.
<i>LAG_TA</i>	= Firm <i>i</i> 's total assets (#6) at year-end <i>t-1</i> .
<i>MNC</i>	= 1 if firm <i>i</i> is a multinational corporation (i.e., if the absolute value of #273 or #64 is greater than zero) in year <i>t</i> , and 0 otherwise.
<i>BLOAT</i>	= Firm <i>i</i> 's extent of balance sheet bloat, measured as shareholders' equity (#216) minus cash and short-term investments (#1), plus total debt (#9 + #34) in year <i>t-1</i> , scaled by net sales (#12) at the end of year <i>t-2</i> .
<i>LAG_NDTL</i>	= Firm <i>i</i> 's net deferred tax liability (i.e., deferred tax liabilities minus net deferred tax assets) at year-end <i>t-1</i> , scaled by total assets at year-end <i>t-2</i> .
<i>SC_PT_EM</i>	= Firm <i>i</i> 's year <i>t</i> pretax earnings management (originally reported minus restated pretax net income) scaled by total assets at year-end <i>t-1</i> .
<i>INV_MILLS</i>	= Inverse Mills ratio from the first-stage of the Heckman (1979) two-step estimation.
$\Sigma_j INDUS$	= 1 (0) if firm <i>i</i> is (is not) in industry <i>j</i> in year <i>t</i> , based on 3-digit SIC codes.

Variables Used in Equation (7):

<i>RESTATE</i>	= 1 if firm <i>i</i> appears in the GAO (2002) restatement report and restated its earnings downward (i.e., had income-increasing earnings management) for year <i>t</i> , and 0 otherwise.
<i>BT_DIFF</i>	= Firm <i>i</i> 's book-tax differences, which equal book income less taxable income scaled by lagged total assets. Book income is pre-tax income (<i>Compustat</i> #170) in year <i>t</i> . Taxable income is calculated by grossing-up the sum of current federal tax expense (#63) and current foreign tax expense (#64) and subtracting the change in NOL carryforwards (#52) in year <i>t</i> . If current federal tax expense is missing, total current tax expense is calculated by subtracting deferred taxes (#50), state income taxes (#173) and other income taxes (#211) from total income taxes (#16) in year <i>t</i> .
<i>MJAC</i>	= Firm <i>i</i> 's abnormal accruals in year <i>t</i> computed using the modified Jones model (Dechow et al. 1995).
<i>MBE</i>	= 1 if the firm <i>i</i> meets or beats the consensus annual <i>I/B/E/S</i> analyst earnings forecast in year <i>t</i> , and 0 otherwise, unless the firm has no analyst following, in which case the dummy variable is 1 if current period earnings are the same as, or higher than, the prior year's earnings (#18).
<i>FIN_R</i>	= Firm <i>i</i> 's external financing raised, equals the sum of additional cash raised from the issuance of common and preferred stock (#108) and the issuance of long-term debt (#111) during year <i>t</i> , scaled by average total assets for year <i>t</i> .
<i>LEV</i>	= Firm <i>i</i> 's leverage ratio, calculated as short term debt (#34) plus long term debt (#9) at year-end <i>t-1</i> , scaled by total assets at year-end <i>t-1</i> .
<i>INV_INTCOV</i>	= The inverse of firm <i>i</i> 's interest coverage ratio, calculated as interest expense (<i>Compustat</i> #15) in year <i>t</i> , divided by operating income before depreciation (<i>Compustat</i> #13) in year <i>t</i> .
<i>EPS_G</i>	= Firm <i>i</i> 's earnings per share (EPS) growth, equal to the number of consecutive quarters (up to eight) of growth in EPS going back in time from quarter 4 of the year prior to year <i>t</i> , where $EPS_Q4_t - EPS_Q4_{t-1}$ is EPS growth.

<i>STRING</i>	= Firm <i>i</i> 's number of consecutive quarters (up to eight) with positive forecast errors going back in time from quarter 4 of the year prior to year <i>t</i> , where forecast error is actual earnings minus the most recent consensus analyst forecast from <i>I/B/E/S</i> , unless the firm has no analyst following, in which case we consider whether current period earnings are the same as, or higher than, the earnings in the same quarter of the prior year (#18).
<i>CFO</i>	= Firm <i>i</i> 's cash flows from continuing operations (#308 - #124) in year <i>t</i> , scaled by total assets at year-end <i>t-1</i> .
<i>ΔCFO</i>	= Firm <i>i</i> 's change in cash flows from continuing operations from year <i>t-1</i> to <i>t</i> , scaled by total assets at year-end <i>t-1</i> .
<i>ΔAR</i>	= Firm <i>i</i> 's change in accounts receivable (#2), scaled by total assets at year-end <i>t-1</i> .
<i>ΔINV</i>	= Firm <i>i</i> 's change in inventories (#3), scaled by total assets at year-end <i>t-1</i> .
<i>ΔCASHSALES</i>	= Firm <i>i</i> 's percentage change in cash sales equals net sales (#12) less change in accounts receivable (#2) for year <i>t</i> minus net sales less change in accounts receivable for year <i>t-1</i> , scaled by net sales less change in accounts receivable for year <i>t-1</i> .
<i>ΔEARN</i>	= Firm <i>i</i> 's earnings (#18) from year <i>t</i> , scaled by total assets at year-end <i>t-1</i> , less earnings (#18) in year <i>t-1</i> , scaled by total assets at year-end <i>t-1</i> .
<i>ABNΔEMPLOYEES</i>	= Firm <i>i</i> 's abnormal change in employees equals the number of employees (#29) for year <i>t</i> minus the number of employees for year <i>t-1</i> , scaled by the number of employees for year <i>t-1</i> , less total assets (#6) at year-end <i>t</i> minus total assets at year-end <i>t-1</i> , scaled by total assets at year-end <i>t-1</i> .
<i>OP_LEASES</i>	= 1 if firm <i>i</i> 's future operating lease obligations (#96, #164, #165, #166 or #167) are greater than zero, and 0 otherwise.

FIGURE 2
Distributions of *C_RATE* and *NC_RATE*



See Figure 1 for complete variable definitions.